

HINGE FIELDS

Executive identity and a calculus
of consequential boundaries

Signed distance. Spatial axes. Temporal phase.
Contact. Release. Transport. State transition.

Tom Klootwijk

Principal Transdisciplinary Technologist
Computational Naturalist

NL200678942 / 10-07-1990

Computational naturalism, cognition, and creative systems design.

The integrated proposition

Investigate what a system can do, locate the boundary at which its available actions or material relations change, and build the computation that makes that change inspectable.

This is the connection between Tom Klootwijk’s professional title, practice identity and hinge framework. **Principal Transdisciplinary Technologist** names responsibility for the technical direction across disciplines. **Computational Naturalist** names the investigative practice: observe mechanisms in living, cognitive and engineered systems, express their relevant structure computationally, and develop models, instruments and experiments from that expression.

The executive vision is to build a mechanism-centred practice whose outputs are not just descriptions but usable objects: an interface reconstruction, an event definition, a measured trajectory, a simulation, a test, a visual instrument or an invention. Boundaries and transitions provide the organising question. Biology supplies physical cases; mathematics supplies representations; engineering supplies implementations; cognition supplies questions about sensing, inference and action.

The signed-distance field supplies a precise *where*. The hinge axis supplies a constrained *how*. Temporal phase supplies a *when within a cycle*. A constitutive law or feedback rule supplies the relevant *coupling*. An event condition and reset supply the *before and after*. These objects belong together, but they are not interchangeable.

The expanded formulation is:

geometry + motion + coupling + history → event + changed future conditions.

The most important addition to the earlier *Hinge Events* atlas is a full passage from a static border to a dynamical mechanism. That passage includes distance gradients, contact Jacobians, rotation matrices, temporal phase, constrained dynamics, impacts, accumulated state, event-time sensitivity, numerical representation and computational verification.

How this edition is grounded. The newest 41-page project dialogue introduces signed distance as “root and stem” on pages 37–39 and spatial–temporal hinge axes on pages 39–41. The earlier 35-page dialogue supplies the $\phi/\Psi/\Delta/T$ vocabulary. The professional role and practice definition are supplied in the accompanying request. These are the project’s source materials. Standard methods are attributed in the references; the specific integrations, proposed models and synthetic benchmarks in this edition are developed here. Source-to-model changes are recorded in Appendix A. [U1]; [U2]; [U4].

Reading routes. Chapters 1 and 12 form the executive presentation. Chapters 2–7 contain the mathematical expansion. Chapters 8–9 connect it to physical, cognitive and engineered cases. Chapters 10–11 specify implementation and worked calculations. The standalone Python companion executes twelve verification tests and reproduces the benchmark values.

Contents

The integrated proposition	i
1 Identity, job description and executive vision	1
1.1 Professional identity	1
1.2 Integrated job description	1
1.3 Executive vision	2
1.4 What counts as an output	2
2 Root, stem and the typed hinge vocabulary	4
2.1 Preserve the source; make the variables operational	4
2.2 Root: the domain and its boundary	5
2.3 Stem: the processes supported by that geometry	5
2.4 Three meanings of hinge	5
3 Signed distance: exact geometry of the border	6
3.1 The signed-distance definition	6
3.2 The Eikonal property and what it actually says	6
3.3 Normals, closest points and curvature	6
3.4 Where a sharp border is not differentiable	7
3.5 Exact primitives and useful non-distance functions	8
3.6 Transformations: what preserves the metric	8
3.7 Combining bodies without confusing the field with the shape	8
4 Spatial hinges and temporal phase	10
4.1 A physical axis and its rotation	10
4.2 The homogeneous matrix is spatial, not a hidden clock	10
4.3 From hinge angle to normal approach speed	10
4.4 Two-body motion and a moving border	11
4.5 Temporal phase: the lower-case psi	12
4.6 Multiple hinges and noncommuting motion	12
5 Contact, load and the physical consequence	13
5.1 A gap is a condition, not yet a force	13
5.2 The contact Jacobian converts force to torque	13
5.3 Within-mode dynamics	14
5.4 Impact: a velocity reset at a boundary	14
5.5 Deformation, friction and retention	14
5.6 Fracture, dose and endpoints after contact	15
5.7 The energy ledger	15
6 Hinge events, memory and sensitivity	16
6.1 The continuous state and the discrete mode	16
6.2 The bit is a report about a state	16
6.3 Crossing, grazing, touch and passage	16
6.4 Hysteresis and finite dwell time	17
6.5 A contact can change the future without changing position	17
6.6 Event time is a sensitive output	18
6.7 The saltation matrix: sensitivity across the hinge	18
6.8 Bifurcation is a separate question	18
7 Space–time, margins and measured uncertainty	19
7.1 A moving boundary is a world tube	19
7.2 Spatial SDF versus a distance in space–time	19
7.3 Distance, residual and probability are different margins	19
7.4 Spatial uncertainty becomes timing uncertainty	20
7.5 Sample timing and the delta–delta calculation	20
7.6 Reachability is more informative than closeness	21

8	Physical hinge mechanisms	22
8.1	Intermeshing gears: <i>Issus</i> nymphs	22
8.2	Trap-jaw ants: the same rotation, different external contact	22
8.3	Bombardier beetles: pressure feedback moves the boundary	23
8.4	Sharpshooter excretion: a deformable droplet on a moving stylus	23
8.5	Woodpeckers: contact as an impulse-transmission problem	24
8.6	Penguin droppings: discharge, then trajectory	24
8.7	Duck gut passage: an exit is not a terminal state	25
8.8	Seabird guano: local deposition, regional atmospheric consequences	26
8.9	Fungal occupation of ant tissues: location before control claims	26
8.10	Plant volatiles: a chemical boundary, not a radio command	26
8.11	Rainbow optics: the material surface and the ray-map fold	27
9	Cognition, creative systems and engineered hinges	28
9.1	The object of the cognitive investigation	28
9.2	A local decision boundary can produce a bit	28
9.3	Extended reality: a visible boundary with a measurable delay	28
9.4	Bugfinding as locating a broken explanatory link	29
9.5	Representation and the executive vision	29
10	A computational specification for the hinge engine	30
10.1	Separate geometry, dynamics and event services	30
10.2	Choose a representation for a specified purpose	30
10.3	Keep the zero set and the distance property distinct	31
10.4	Conservative stepping needs an actual bound	31
10.5	Locate the event before applying the reset	32
10.6	Record an event as a reproducible object	32
10.7	The supplied computational companion	33
11	Worked calculations and verification results	34
11.1	A rounded tip rotating toward a plane	34
11.2	An approaching point and a moving, expanding sphere	35
11.3	A grazing event that endpoint signs do not reveal	36
11.4	Capillary timing and a ballistic discharge	36
11.5	One threshold versus hysteresis	36
11.6	Saltation tested against an entire trajectory	37
11.7	Verification record	37
12	Executive program: from observation to invention	38
12.1	The executive mandate	38
12.2	Three coordinated workstreams	38
12.3	A sequence of acceptance decisions	39
12.4	What the practice delivers	39
12.5	An operational exACTiom for the framework	39
12.6	Final professional statement	40
A	Source-to-model crosswalk	41
A.1	Project source register	41
A.2	What is retained, specified or changed	41
B	A reusable hinge dossier	43
B.1	The record to complete for a new mechanism	43
B.2	A compact review question	43
	Research and technical references	44

Identity, job description and executive vision

1.1 Professional identity

Professional presentation

Tom Klootwijk

Principal Transdisciplinary Technologist

Computational Naturalist

Computational naturalism, cognition, and creative systems design.

Identity record: NL200678942 / 10-07-1990.

The title names a function rather than a catalogue of subjects. The recurring function is to find mechanisms, decide how to represent them, connect the required methods, and turn that representation into working technical output. In this document, “principal” locates technical direction and accountability; “transdisciplinary” locates integration across methods; “technologist” locates the work in construction as well as investigation.

Computational naturalism is the practice name. It includes the field observation that begins with an insect on a plant, the geometric reconstruction that establishes a boundary, the calculation that predicts a contact, and the instrument that compares the prediction with what actually happened. It also includes the reverse route: an unexpected computational failure can suggest a new experiment or reveal which observation was missing.

The same identity accommodates creative work without making creativity a substitute for mechanism. A spatial installation, a simulation or an extended-reality interface can make a process inspectable. Its role is strongest when the viewer can change a parameter and see what the model actually predicts: a hinge closes sooner, a contact patch shifts, a plume reaches a receiver later, or a numerical event disappears when the timestep changes.

1.2 Integrated job description

First-person statement

I investigate how living, cognitive and engineered systems behave, particularly at boundaries, transitions and failure points. I combine observation, computational modelling and creative technical experimentation to identify mechanisms and develop models, tools, interfaces and inventions. As Principal Transdisciplinary Technologist, I define the technical direction, connect the necessary methods and take responsibility for the relationship between what is observed, what is computed and what is built.

The work begins by selecting the unit of explanation. Is the object a rotating joint, a deforming interface, a sensory response, an accumulated exposure, a sequence of contacts or a distributed exchange? Selecting that object determines what must be measured. A useful job description therefore contains responsibility

for *problem formulation*, not only for solving a problem already supplied by someone else.

The next responsibility is translation between representations. An image is not yet a three-dimensional body. A three-dimensional body is not yet a signed-distance function. A signed-distance function is not yet a contact law. A contact law is not yet a prediction of retention, release or failure. Moving from one representation to another requires a stated transformation and a check that the relevant information survived it.

The principal role also owns integration decisions. It decides when a detailed model is necessary and when a reduced model is adequate. It chooses the interface between geometry and dynamics, the event definition used across software components, the reference frame shared by sensors, and the residuals that determine whether a prototype is behaving as specified. The purpose is not to maximise technical vocabulary. It is to keep a chain of explanation intact from observation to output.

Finally, the role owns the failure investigation. A failed prediction is examined at the layer where it could originate: missing geometry, wrong material law, timing offset, unobserved state, numerical approximation or an incorrect event definition. “Bugfinding” then names a general investigative habit: locating the difference between expected and actual behaviour, rather than assuming that every mismatch is a software defect.

1.3 Executive vision

First-person executive vision

My aim is to build a practice that makes mechanisms legible and operational. I treat an interface as a place where matter, force, energy or information can be coupled; a hinge as a constrained direction of change; and an event as a transition that alters what can happen next. I use computation to expose those relations, not to replace them with an attractive picture.

The output I seek is a connected chain: observation, geometry, dynamics, event, consequence and implementation. Each project should leave behind a reusable technical object and a sharper explanation of the system it addresses.

This vision unifies breadth through a common workflow. It does not require a bird, an insect, a person and a processor to possess the same internal architecture. Instead, it asks whether an explicit description can identify the relevant bodies or states, the interfaces between them, the coupling across those interfaces, and the transitions that make a difference to future behaviour.

Three production streams follow from this executive direction. The **interface observatory** records geometry and trajectories. The **hinge engine** represents motion, coupling and events. The **mechanism studio** turns those representations into simulations, instruments and interactive technical artefacts. These are proposed workstreams for the practice, not separate professional identities.

A project’s executive summary should answer five compact questions in prose: what changes; where and when it changes; what physically or computationally couples the parts; how the change is detected; and what becomes different afterward. The mathematics in the following chapters supplies the technical content behind those questions.

1.4 What counts as an output

A mechanism dossier becomes a useful output when it has a stable identity, a defined scope and a reproducible chain of evidence. For a hinged contact, that might mean a calibrated geometry file, a local coordinate

frame, a motion trajectory, a force or material model, an event record and a benchmark. For an interface in extended reality, it might mean a measured coordinate transform, a latency model, a geometric threshold and a reproducible interaction test.

A collection of such dossiers can become a library, but only if the entries share an interface. The shared interface is not a claim of universal anatomy. It is a data and reasoning contract: units are declared; states are typed; events have conditions; a reset has a defined target; and an output can be traced to inputs. This is how an executive vision becomes a technical operating system for the work.

Operational consequence. The identity names the practitioner. The title names technical responsibility. Computational naturalism names the investigative method. The hinge framework names an organising architecture for models and experiments.

Root, stem and the typed hinge vocabulary

2.1 Preserve the source; make the variables operational

The latest source uses the signed-distance field as “root and stem”, gives $f < 0$, $f = 0$ and $f > 0$ symbolic roles, and introduces lower-case ϕ and ψ as spatial and temporal axes. It asks for a transformation $\mathbf{M}(\phi, \psi)$ without specifying its entries. These are the immediate starting points for this expansion. [U1, pp. 37–41].

The technical construction here keeps that spatial–temporal organisation and assigns each object one mathematical type. A scalar distance is not an angle; an angle is not an axis vector; an axis vector is not a matrix; and a local binary state is not the complete state of a mechanism. The separation is useful because it lets the symbols interact without concealing changes of units or meaning.

Symbol	Type and unit	Operational meaning
$d(\mathbf{x}, t)$	Scalar, length	Signed spatial distance to a specified boundary.
$\Gamma(t)$	Surface or curve	The zero set $d = 0$.
ϕ	Scalar angle, radians	A spatial hinge coordinate.
$\mathbf{e}$	Unit vector	Direction of a physical rotation axis.
ψ	Scalar phase, radians	Position within a specified temporal cycle.
ω_ψ	Phase rate, rad/s	Rate at which the temporal cycle advances.
$\mathbf{q}$	Configuration vector	Positions, joint angles and other geometric coordinates.
$\mathbf{z}$	Continuous state	Configuration, velocity and needed internal variables.
m	Discrete mode label	Separated, contacting, attached, released, and so on.
Ψ_i	Local bit, 0 or 1	Encoding of one declared two-state property.
$g_i(\mathbf{z}, t)$	Scalar event residual	Condition for a particular event; units depend on the event.
Δ	Difference operator	Change over a declared sampling interval.
$\mathcal{T}$	Geometric template family	Sphere, capsule, cone, box or measured geometry.
$\mathbf{T}$	Homogeneous transform	Rotation and translation of a rigid body.
$\mathcal{R}$	Reset map	Change in state associated with a defined event.

This edition reserves d for signed distance, ϕ for the spatial angle, and ψ for temporal phase. Where an actual axis must be named, it is a vector such as $\mathbf{e}_\phi$. A phase need not have a spatial axis. When a physical mechanism contains a second spatial hinge, its axis can be $\mathbf{e}_2$, with a separate joint angle. This prevents an abstract clock coordinate from becoming a physical direction merely through notation.

2.2 Root: the domain and its boundary

The root of a model is its declared domain: the body, region or state set being represented. A jaw model might use a solid cuticular body; a droplet model a liquid region; a navigation model the space not occupied by obstacles. In each case, the meaning of “inside” is supplied before calculating a sign.

This differs from using inside and outside as universal labels for truth and simulation. A virtual reconstruction can model a real interior correctly; an exterior point can be ordinary physical air. The distance sign classifies location relative to a chosen region. A project may attach an additional interpretation to the classification, but that interpretation is a separate mapping.

The domain declaration should include scale. A microscopically rough cuticle can be represented as smooth at a larger resolution. The resulting boundary is a model at that resolution. Shrinking the voxel size does not automatically recover unmeasured structure; adding a sharper colour edge does not add geometric information.

2.3 Stem: the processes supported by that geometry

The stem is the connected sequence of operations that grows from the domain: transformation, motion, interaction, event and continuation. For a jaw, the sequence could be rotation about an axis, decreasing clearance, first contact, increasing load, slip or puncture, and release. For a droplet, it could be volume accumulation, deformation, neck thinning, detachment and flight.

A useful formal mechanism record is

$$\mathcal{H}_i = (\Omega_i, d_i, \mathbf{q}_i, \mathbf{z}_i, m_i, \mathbf{f}_i, g_i, \mathcal{R}_i, C_i), \quad (2.1)$$

where $\mathbf{f}_i$ is within-mode dynamics and C_i contains the coupling and constitutive parameters required by that mechanism. This record is a construction for this framework. It packages standard geometric and dynamical objects in a form that can be used consistently across the physical atlas.

Not every mechanism needs every field. A static clearance test needs geometry but not a temperature state. A pressure valve needs pressure even when its external body does not translate. The record should become smaller when the mechanism permits it, not larger merely to appear universal.

2.4 Three meanings of hinge

A **kinematic hinge** is a physical constraint that permits particular relative motion, typically one rotational degree of freedom. A **hybrid hinge event** is a transition between modes, such as separation and contact. A **dynamical hinge** is a change in the structure of solutions, such as loss of stability or a bifurcation as a parameter varies.

These can coincide. A physical hinge can carry a latch; releasing the latch changes the mode; varying stiffness can change stability. They can also occur independently. A droplet may detach without a revolute joint. A numerical classifier may change output without a physical discontinuity. A rotating joint can move continuously for hours without a change of mode.

Operational consequence. “Hinge” becomes a precise term when the model states whether it refers to a physical constraint, an event rule or a change in dynamical structure.

Signed distance: exact geometry of the border

3.1 The signed-distance definition

Let $\Omega \subset \mathbb{R}^3$ be a regular solid region with boundary $\Gamma = \partial\Omega$. At a point $\mathbf{x}$, define

$$d(\mathbf{x}) = \begin{cases} -\inf_{\mathbf{y} \in \Gamma} \|\mathbf{x} - \mathbf{y}\|, & \mathbf{x} \in \text{int } \Omega, \\ 0, & \mathbf{x} \in \Gamma, \\ +\inf_{\mathbf{y} \in \Gamma} \|\mathbf{x} - \mathbf{y}\|, & \mathbf{x} \notin \Omega. \end{cases} \quad (3.1)$$

The sign classifies side; the magnitude gives shortest Euclidean distance. The zero set identifies the boundary. Level-set methods represent interfaces through zero sets, while distance-based methods exploit the metric information as well. These are related uses of implicit geometry, not identical requirements. [1, 2]

The sign convention is chosen here as negative inside and positive outside. Reversing it changes normals and inequality conventions but not the underlying shape. A software system must choose one convention and apply it consistently to surface extraction, normal calculation, collision detection and force directions.

3.2 The Eikonal property and what it actually says

Where an exact signed-distance field is differentiable,

$$\|\nabla d\| = 1. \quad (3.2)$$

A local unit displacement in the direction normal to a smooth boundary changes the distance by one unit to first order. The equation fixes a spatial rate of change. It does not specify energy, mass, information or the volume of the enclosed region.

A direct counterexample makes the distinction exact. Let the region be a sphere of fixed radius R :

$$d(\mathbf{x}) = \|\mathbf{x}\| - R, \quad V = \frac{4\pi}{3}R^3. \quad (3.3)$$

Move the query point outward from radius $2R$ to radius $3R$. Its distance changes from R to $2R$; the sphere's volume remains $4\pi R^3/3$. The Eikonal property holds away from the centre throughout. Increased exterior distance does not force decreased interior volume.

If the radius itself varies, then $\dot{V} = 4\pi R^2 \dot{R}$. That is a geometric volume-change identity. To determine $\dot{R}$ physically requires a motion, growth or transport law. This is the exact replacement for the source dialogue's proposed inference from unit gradient to a zero-sum exchange between symbolic and biological quantities. [U1, pp. 38–40].

3.3 Normals, closest points and curvature

At a smooth point with a unique closest boundary point, the outward normal is

$$\mathbf{n} = \nabla d, \quad \pi(\mathbf{x}) = \mathbf{x} - d(\mathbf{x})\nabla d(\mathbf{x}). \quad (3.4)$$

Here π returns the closest point on the surface. The expression is valid within a region where that closest point and normal are well-defined. It is useful for locating a contact patch, constructing an offset surface, or turning a point query into a local frame.

For a differentiable implicit function F with the same zero set and $\nabla F \neq 0$ at the query point, the normal is instead

$$\mathbf{n} = \frac{\nabla F}{\|\nabla F\|}.$$

The division matters. A function can locate a boundary correctly and still give a gradient with the wrong magnitude for a distance calculation.

With outward normal convention, define the sum of principal curvatures by

$$\kappa = \nabla \cdot \mathbf{n}. \quad (3.5)$$

For a sphere at its surface, $\kappa = 2/R$. For the exact spherical distance field away from the centre,

$$\nabla^2 d = \frac{1}{r} (\mathbf{I} - \mathbf{n}\mathbf{n}^\top), \quad r = \|\mathbf{x}\|. \quad (3.6)$$

The Hessian has zero curvature in the radial direction and $1/r$ in each tangent direction. Thus a field contains more than an inside/outside bit: its derivatives carry orientation and curvature information relevant to contact, wetting and numerical conditioning.

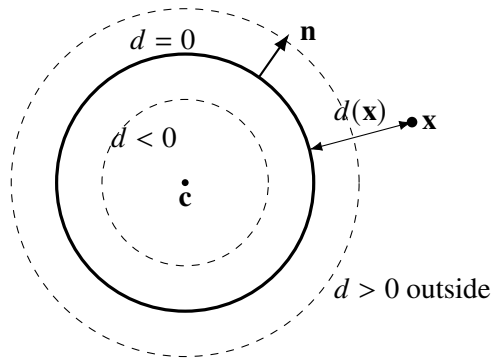


Figure 3.1: A spatial distance field and its offsets. The normal and distance are geometric quantities; a force or a transition rule must be supplied separately. Original schematic.

3.4 Where a sharp border is not differentiable

At some points there are multiple nearest surface points. The centre of a sphere is the simplest example: every point on its surface is equally near. Between two opposing walls, the midline can likewise have two closest points. At such places the distance is meaningful but a unique classical gradient may not exist.

This set is important for hinge analysis. A closest point can jump from one tooth to another, from a jaw tip to its flank, or from a droplet's main body to its neck. A numerical gradient that averages the competing directions can look smooth while representing neither actual contact normal. The model should carry the identity of the closest feature or use a contact representation that handles multiple candidates.

Corners have the same practical consequence at the boundary. An ideal cone apex has no single smooth surface normal. A physical tip has a finite radius or smaller-scale structure. The source's cone, sphere and pyramid are useful templates, but local contact mechanics depends strongly on the features those templates simplify.

3.5 Exact primitives and useful non-distance functions

For a sphere centred at $\mathbf{c}$,

$$d_{\text{sphere}}(\mathbf{x}) = \|\mathbf{x} - \mathbf{c}\| - R. \quad (3.7)$$

For a plane with unit normal $\mathbf{n}_0$ and point $\mathbf{a}$,

$$d_{\text{plane}}(\mathbf{x}) = \mathbf{n}_0 \cdot (\mathbf{x} - \mathbf{a}). \quad (3.8)$$

For a capsule of radius r around a segment from $\mathbf{a}$ to $\mathbf{b}$, with $\mathbf{a} \neq \mathbf{b}$, calculate

$$s = \text{clamp} \left(\frac{(\mathbf{x} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a})}{\|\mathbf{b} - \mathbf{a}\|^2}, 0, 1 \right), \quad (3.9)$$

$$d_{\text{capsule}}(\mathbf{x}) = \|\mathbf{x} - \mathbf{a} - s(\mathbf{b} - \mathbf{a})\| - r. \quad (3.10)$$

This is useful for a first model of a slender leg, stylus or rod with rounded ends. It is exact for the capsule, not for every appendage represented by it.

For a box centred at the origin with positive half-extents $\mathbf{b}$, let $\mathbf{q} = |\mathbf{x}| - \mathbf{b}$, with componentwise absolute value. Then

$$d_{\text{box}}(\mathbf{x}) = \|\max(\mathbf{q}, 0)\| + \min(\max(q_1, q_2, q_3), 0). \quad (3.11)$$

The formula preserves diagonal distance outside a corner. Replacing it with only the largest coordinate excess would preserve a boundary but not Euclidean distance.

By contrast, $F(\mathbf{x}) = \|\mathbf{x} - \mathbf{c}\|^2 - R^2$ has the correct spherical zero set but units of length squared. Its gradient magnitude is $2\|\mathbf{x} - \mathbf{c}\|$, not one. It remains a valid implicit function. The error arises only when it is used as though it were a distance.

3.6 Transformations: what preserves the metric

For a rigid transformation $\mathbf{x} = \mathbf{R}\mathbf{x} + \mathbf{p}$ with $\mathbf{R}^T\mathbf{R} = \mathbf{I}$, the transformed field is

$$d(\mathbf{x}) = d_0(\mathbf{R}^T(\mathbf{x} - \mathbf{p})). \quad (3.12)$$

It remains an exact Euclidean SDF because a rotation and translation preserve distances. Uniform positive scaling by s gives

$$d(\mathbf{x}) = s d_0 \left(\frac{\mathbf{R}^T(\mathbf{x} - \mathbf{p})}{s} \right).$$

The factor s restores distance units in world coordinates.

For a general invertible deformation $\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{p}$, the pullback $F = d_0(\mathbf{A}^{-1}(\mathbf{x} - \mathbf{p}))$ preserves the transformed zero set but generally not distances. Its gradient is $\mathbf{A}^{-T}\nabla d_0$. Unless the deformation is an isometry or is handled with an appropriate metric, different directions are stretched differently.

A local correction $F/\|\nabla F\|$ can approximate distance close to a regular zero set. It is not a global exact re-distancing operation. A thin appendage stretched into an ellipse must therefore be re-distanced or queried with an actual nearest-point calculation when accurate clearance matters.

3.7 Combining bodies without confusing the field with the shape

For negative-inside implicit fields, minimum and maximum operations encode useful set combinations:

$$F_{A \cup B} = \min(d_A, d_B), \quad (3.13)$$

$$F_{A \cap B} = \max(d_A, d_B), \quad (3.14)$$

$$F_{A \setminus B} = \max(d_A, -d_B). \quad (3.15)$$

They provide the expected membership logic under ordinary nondegenerate geometry, but the resulting values need not be exact Euclidean distances on both sides. The distinction matters most around overlaps, concavities and competing closest features.

A smooth minimum, for example

$$\text{smin}_{\beta}(a, b) = -\frac{1}{\beta} \log \left(e^{-\beta a} + e^{-\beta b} \right), \quad (3.16)$$

with β in inverse length, also changes geometry. At $a = b = 0$, it equals $-\log 2/\beta$, not zero. Smoothing is therefore not merely cosmetic. It can close a clearance, round a tip or move an event time. Its geometric effect must be included in the model error rather than described as an unchanged exact border.

Operational consequence. The zero set determines the represented border. Exact distance, differentiable normals, curvature and distance-preserving transformations are additional properties that must each be checked.

Spatial hinges and temporal phase

4.1 A physical axis and its rotation

Let a hinge axis pass through $\mathbf{a}$ in unit direction $\mathbf{e}$. A point with reference position $\mathbf{X}$ moves as

$$\mathbf{x}(\phi) = \mathbf{a} + \mathbf{R}(\mathbf{e}, \phi)(\mathbf{X} - \mathbf{a}). \quad (4.1)$$

Define the skew-symmetric matrix $[\mathbf{e}]_{\times}$ by $[\mathbf{e}]_{\times} \mathbf{v} = \mathbf{e} \times \mathbf{v}$. The rotation is

$$\mathbf{R} = \mathbf{I} + \sin \phi [\mathbf{e}]_{\times} + (1 - \cos \phi) [\mathbf{e}]_{\times}^2. \quad (4.2)$$

This is Rodrigues' formula. It supplies the explicit spatial transformation absent from the source's unspecified matrix. Its geometric interpretation follows the axis-angle representation used in rigid-body robotics. [3]

For the rotation matrix, $\mathbf{R}\mathbf{e} = \mathbf{e}$. The hinge direction is therefore an eigenvector with eigenvalue one. The angle ϕ is not that eigenvector: it determines how far other directions rotate around it. At a zero rotation every direction is fixed, so extracting a unique axis from the matrix alone becomes indeterminate. A mechanism's known anatomical or engineered axis can still be specified independently.

This gives a precise reading of "eigenvector axis": it is a direction left unchanged by the corresponding rotation. It does not imply that every variable with a Greek symbol is an eigenvector, nor that an arbitrary transformation matrix describes a physical hinge.

4.2 The homogeneous matrix is spatial, not a hidden clock

A rigid pose can be written

$$\begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}^T & 1 \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} \mathbf{X} \\ 1 \end{bmatrix}. \quad (4.3)$$

The final coordinate is the homogeneous constant one, not elapsed time. Time determines the pose through $\mathbf{R}(t)$ and $\mathbf{p}(t)$; it does not take the place of the final one. This distinction matters when translating the source's (x, y, z, t) notation into actual mechanics. [U1, p. 40].

For a hinged body whose reference field is d_0 , the world-space field is obtained by the inverse pose:

$$d_B(\mathbf{x}, \phi) = d_0 \left(\mathbf{R}(\phi)^T (\mathbf{x} - \mathbf{p}(\phi)) \right). \quad (4.4)$$

A moving rigid jaw can therefore carry a fully defined field rather than a moving drawn outline. Every query point returns its clearance to the current body, and the returned normal rotates with the body.

4.3 From hinge angle to normal approach speed

Differentiate the point position with respect to its angle:

$$\frac{\partial \mathbf{x}}{\partial \phi} = \mathbf{e} \times (\mathbf{x} - \mathbf{a}), \quad \dot{\mathbf{x}} = \dot{\phi} \mathbf{e} \times (\mathbf{x} - \mathbf{a}) \quad (4.5)$$

for a fixed axis in a fixed base. A point farther from the axis has a larger linear speed at the same angular speed. Only the component normal to a target closes its gap.

For a point moving relative to a stationary target with SDF d_T , define $g(\phi) = d_T(\mathbf{x}(\phi))$. Then

$$\frac{\partial g}{\partial \phi} = \nabla d_T \cdot [\mathbf{e} \times (\mathbf{x} - \mathbf{a})] = \mathbf{e} \cdot [(\mathbf{x} - \mathbf{a}) \times \mathbf{n}]. \quad (4.6)$$

This derivative is the normal contact Jacobian for that point and hinge. It measures how much clearance changes per radian. It is simultaneously related to the normal force's moment arm about the hinge axis. Geometry therefore connects approach speed and torque transmission through the same scalar.

When the Jacobian is nearly zero, angular motion is locally tangent to the target. Large angular speed can then produce little normal approach. Conversely, a small force at a long normal moment arm can produce appreciable torque. An “apex” alone does not determine either quantity; orientation and leverage must be included.

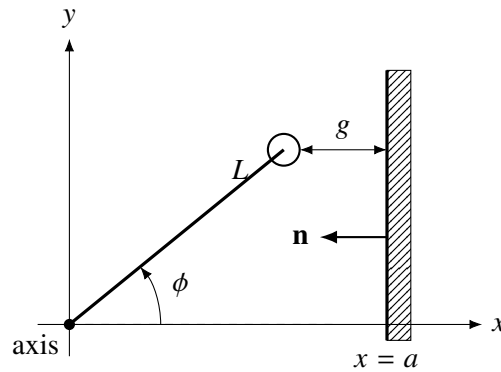


Figure 4.1: A planar hinge approaches a half-space. For a rounded tip, $g = a - L \cos \phi - r$. Reducing the angle closes the gap on the illustrated branch. Original benchmark schematic, not a measured insect joint.

4.4 Two-body motion and a moving border

For a moving point $\mathbf{x}_A(t)$ and a moving target field $d_B(\mathbf{x}, t)$,

$$\dot{g} = \partial_t d_B + \dot{\mathbf{x}}_A \cdot \nabla d_B. \quad (4.7)$$

At a regular boundary point with target-surface velocity $\mathbf{u}_B$, this becomes

$$\dot{g} = \mathbf{n} \cdot (\dot{\mathbf{x}}_A - \mathbf{u}_B). \quad (4.8)$$

A stationary tip can be contacted by an approaching target. A rapidly moving tip can maintain constant clearance if the target moves alongside it. Contact is governed by relative normal motion, not by the speed of one object viewed alone.

For rigid-body velocity,

$$\mathbf{u}_B = \dot{\mathbf{p}}_B + \boldsymbol{\omega}_B \times (\mathbf{x} - \mathbf{p}_B).$$

Rotation of the target therefore matters even if its centre is fixed. An insect's rotating leg, an opening bird beak and a turning mechanical gate all require the velocity of the actual surface feature, not merely the velocity of a body centroid.

For a fully deforming interface, introduce an implicit function F satisfying

$$\partial_t F + \mathbf{u} \cdot \nabla F = 0. \quad (4.9)$$

On the zero set, its normal speed is $V_n = -\partial_t F / \|\nabla F\|$. Ordinary advection need not preserve the exact SDF property away from the interface. Following the boundary and maintaining a globally exact distance are separate numerical tasks. [1, 4]

4.5 Temporal phase: the lower-case psi

Let ψ describe position within a specific cycle. A phase oscillator can be written

$$\dot{\psi} = \omega_\psi(\mathbf{z}, t), \quad \psi \pmod{2\pi}. \quad (4.10)$$

An angular hinge trajectory can then depend on phase:

$$\phi(\psi) = \phi_0 + A \sin \psi, \quad \dot{\phi} = A \cos \psi \dot{\psi}. \quad (4.11)$$

Here ϕ sets spatial orientation and ψ organises temporal progression. This gives the source's two-symbol distinction a concrete implementation. It does not impose a particular frequency on every organism, and it does not require ψ to be a vector in physical space.

The second derivative shows why phase and geometry should remain distinct:

$$\ddot{\phi} = -A \sin \psi \dot{\psi}^2 + A \cos \psi \ddot{\psi}.$$

Acceleration can change because the hinge is at a different phase, because the phase advances faster, or because the phase rate itself changes. Those are different causes. Writing “time rotation” without this distinction hides them.

For noncyclic events, a phase is unnecessary. Elapsed time since release, a timer s with $\dot{s} = 1$, or an accumulated state can be more appropriate. A droplet travelling away after detachment does not have to be forced into a periodic representation.

4.6 Multiple hinges and noncommuting motion

For a serial chain, a point's pose is built from successive transformations,

$$\mathbf{T}_{0n}(\mathbf{q}) = \mathbf{T}_{01}(q_1) \mathbf{T}_{12}(q_2) \cdots \mathbf{T}_{n-1,n}(q_n). \quad (4.12)$$

Rotations generally do not commute. Rotating a jaw about one axis and then another can produce a different pose from reversing the order. Thus a list of angles is incomplete unless the axes, coordinate frames and multiplication order are defined.

A contact residual now has a row Jacobian

$$\mathbf{J}_g = \frac{\partial g}{\partial \mathbf{q}}, \quad \dot{g} = \mathbf{J}_g \dot{\mathbf{q}} + \partial_t g.$$

Each column measures one coordinate's contribution to the gap. A closed linkage additionally imposes constraints $\mathbf{c}(\mathbf{q}) = 0$ and velocity compatibility $\mathbf{C}\dot{\mathbf{q}} = 0$, with $\mathbf{C} = \partial \mathbf{c} / \partial \mathbf{q}$. Treating every angle as independent would allow impossible motions.

Operational consequence. The precise spatial–temporal pairing is an axis and angle that determine geometry, coupled to a clock or phase law that determines its evolution. Their relationship is specified by the mechanism rather than assumed from their names.

Contact, load and the physical consequence

5.1 A gap is a condition, not yet a force

A model first establishes whether relevant surfaces meet. For a rounded tip of radius r whose centre is outside a target solid, the exterior clearance can be written $g = d_T(\mathbf{x}_c) - r$. For a general body, the model must search appropriate surface features or represent collision in configuration space. A centre-to-centre distance alone is usually not a body-to-body clearance.

Define the forbidden configuration set as configurations in which the bodies' interiors overlap. Its boundary can be used to define contact. The space may contain metres and radians together, so a Euclidean distance in the raw coordinate vector is not automatically meaningful. A local contact residual with units of length can be useful without being an exact signed distance in configuration space.

In an ideal rigid unilateral-contact model,

$$g \geq 0, \quad \lambda_n \geq 0, \quad g\lambda_n = 0. \quad (5.1)$$

Positive separation carries no compressive contact force. At contact, the force can be positive while the bodies do not interpenetrate. These relations express a constraint and its force multiplier, not a binary switch inside an animal. Rigid-contact mechanics provides this mathematical language. [5]

5.2 The contact Jacobian converts force to torque

For configuration $\mathbf{q}$, a normal force produces generalised force

$$\boldsymbol{\tau}_c = \mathbf{J}_g^T \lambda_n. \quad (5.2)$$

The same Jacobian converts coordinate velocity into normal gap velocity. The associated power identity is

$$\dot{\mathbf{q}}^T \boldsymbol{\tau}_c = \lambda_n \mathbf{J}_g \dot{\mathbf{q}}.$$

This is a strong technical expression of a hinge: its geometry determines both the permitted motion and how a load acts on that motion. A small rotation does not have a universal consequence. Its leverage depends on the contact location and normal.

For the planar benchmark $g = a - L \cos \phi - r$, the Jacobian is $J_g = L \sin \phi$. At a quasistatic contact,

$$\tau_{\text{act}} + J_g \lambda_n = 0.$$

A closing actuator torque is negative under this angle convention; a repulsive target force provides positive opening torque. Taking magnitudes gives $\lambda_n = |\tau_{\text{act}}| / |J_g|$ only when other torques and acceleration are negligible.

Near a vanishing moment arm, this reduced expression predicts a large force for a given torque. That is a warning about geometric singularity and model adequacy: finite stiffness, friction, alternate contacts and actual geometry become important. It is not a source of unlimited physical force.

5.3 Within-mode dynamics

A general mechanical model may be written

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \nabla_{\mathbf{q}}U = \boldsymbol{\tau}_{\text{act}} + \mathbf{J}_g^T \lambda_n + \boldsymbol{\tau}_{\text{other}}. \quad (5.3)$$

The mass matrix $\mathbf{M}$ determines inertial response, U is potential energy, and the remaining terms represent actuation, contact and other specified loads. The matrix is not chosen simply to make a desired conclusion follow. Its entries are determined by the geometry, masses and modelling assumptions. [3, 5]

For a single compliant hinge, a reduced equation is

$$I\ddot{\phi} + c_{\phi}\dot{\phi} + k_{\phi}(\phi - \phi_{\text{rest}}) = \tau_{\text{act}} + J_g \lambda_n. \quad (5.4)$$

A latch can hold ϕ approximately fixed while elastic energy is stored. Releasing that constraint permits the energy to enter motion. The release condition, the spring law and the resulting trajectory are distinct parts of the model.

For small undamped oscillations of a multi-hinge system, mode shapes satisfy

$$\mathbf{K}\mathbf{e}_j = \omega_j^2 \mathbf{M}\mathbf{e}_j. \quad (5.5)$$

These eigenvectors live in the chosen generalised-coordinate space. They describe coordinated patterns of motion, not necessarily single physical rotation axes. This is a second, different technical use of “eigenvector” in the framework.

5.4 Impact: a velocity reset at a boundary

During an ideal instantaneous collision with a stationary target, configuration remains continuous but velocity can jump. For a single frictionless normal contact,

$$\dot{\mathbf{q}}^+ = \dot{\mathbf{q}}^- + \mathbf{M}^{-1} \mathbf{J}_g^T P_n, \quad (5.6)$$

$$\dot{g}^+ = -e\dot{g}^-, \quad 0 \leq e \leq 1, \quad (5.7)$$

where P_n is normal impulse and e is a restitution coefficient. For an approaching contact,

$$P_n = -\frac{(1+e)\dot{g}^-}{\mathbf{J}_g \mathbf{M}^{-1} \mathbf{J}_g^T}. \quad (5.8)$$

The denominator is inverse effective mass in the contact-normal direction. It can differ substantially from the inverse total mass because rotation contributes to the contact response.

This formula requires the stated single-contact impact model. Multiple simultaneous contacts can couple impulses. Deforming contacts have finite duration. Adhesion can alter separation. Repeated collisions may need time-stepping rather than independent isolated resets. The computational representation should follow the mechanism rather than force every event into this simplest formula.

5.5 Deformation, friction and retention

A compliant model can permit a deformation proxy $\delta = \max(0, -g)$ and use, for example,

$$\lambda_n = \max(0, k_n \delta + c_n \dot{\delta}).$$

The apparent geometric overlap here represents deformation in a reduced model; it is not the same assumption as perfectly rigid nonpenetration. Calibrating k_n from one loading regime does not automatically calibrate a material over all rates and deformations.

Tangential contact adds another direction. A simple dry-friction description constrains $\|\lambda_t\| \leq \mu\lambda_n$ in sticking, with a sliding law at the limit. A target can therefore be in geometric contact but not retained. Retention requires the available contact forces and geometry to resist escape over a defined interval.

An operational capture event could require contact to persist for duration τ_c while relative displacement remains bounded. That definition makes the endpoint inspectable. It also makes clear why first touch, grasp, puncture and eventual feeding are different events.

5.6 Fracture, dose and endpoints after contact

A material failure condition can be introduced through energy release rate G and resistance G_c , with onset modelled at $G = G_c$ under a stated fracture model. These quantities have units of energy per newly created crack area. A distance value cannot substitute for them.

For a history-dependent endpoint, define an internal variable D through a rate law

$$\dot{D} = \mathcal{F}(\sigma, \dot{\epsilon}, T, c, D), \quad g_D = D - D_c. \quad (5.9)$$

The functional form is chosen and calibrated for the particular mechanism. A stress-driven material damage variable and a chemical exposure integral need not share the same law. Their common architecture is accumulation followed by a defined endpoint.

Consequently, two encounters at the same geometric border can have different outcomes. They may differ in speed, orientation, force duration, surface chemistry, target properties or prior history. A binary contact bit omits those distinctions unless they remain elsewhere in the state.

5.7 The energy ledger

For a driven, damped single hinge,

$$E = \frac{1}{2}I\dot{\phi}^2 + U(\phi), \quad \dot{E} = \tau_{\text{act}}\dot{\phi} - c_{\phi}\dot{\phi}^2 + P_{\text{other}}. \quad (5.10)$$

A rotation matrix preserves geometric lengths. It does not make this mechanical energy constant. Energy accounting comes from a balance law, not from the orthogonality of a coordinate transform. Likewise, a physical organism exchanges matter and energy with its surroundings; the relevant boundary terms must be included.

For an ideal one-degree-of-freedom collision with unchanged inertia and restitution e , post-impact kinetic energy is e^2 times pre-impact kinetic energy. The difference enters deformation, heat, sound or other processes represented only implicitly by the reduced impact model. “Event occurred” does not specify that distribution.

Operational consequence. An SDF locates the encounter. A Jacobian turns geometry into approach and leverage. A dynamical or constitutive law determines force, impulse, deformation and later consequence.

Hinge events, memory and sensitivity

6.1 The continuous state and the discrete mode

Let $\mathbf{z}$ contain the continuous variables needed for the model: configuration, velocity, pressure, concentration, stored energy, damage or a timer. Let m identify the current mode. Between events,

$$\dot{\mathbf{z}} = \mathbf{f}_m(\mathbf{z}, \mathbf{u}, t). \quad (6.1)$$

A transition from m to n has an event residual g_{mn} and reset map $\mathcal{R}_{mn}$:

$$g_{mn}(\mathbf{z}, t) = 0, \quad \mathbf{z}^+ = \mathcal{R}_{mn}(\mathbf{z}^-, t), \quad m^+ = n. \quad (6.2)$$

Hybrid-system methods combine continuous evolution and discrete changes of this kind. [6]

A complete event specification also states crossing direction, required mode and any additional conditions. For contact from separation, $g = 0$ and $\dot{g} < 0$ indicate approach. An opening valve might instead use pressure above a threshold while the valve is closed. An elapsed-time event can occur even if the geometry remains stationary.

The word *guard* is standard technical language for such an event condition. It is a mathematical eligibility test, not an ethical rule. Here it means the set of states at which a specified transition is allowed by the model.

6.2 The bit is a report about a state

A local bit can encode whether a particular condition holds:

$$\Psi_{\text{contact}} = \mathbf{1}\{m \text{ is a contact mode}\}. \quad (6.3)$$

Another bit may encode attachment, a third whether a valve is open. These can change independently. Their values do not uniquely determine pressure, velocity, tissue condition or sensory processing.

A state machine can have two modes while each mode contains a continuous range of states. Conversely, a smooth physical system can be given a binary measurement output by applying a threshold. The binary form may belong to the instrument, the analysis or the physical mode structure. Those possibilities should be recorded rather than collapsed into one claim about underlying machinery.

The source's "1-bit" motif therefore becomes useful as a *local encoding*. It can provide compact event logs and control logic without discarding the continuous state that explains the transition. [U2, pp. 1–13].

6.3 Crossing, grazing, touch and passage

A transverse event satisfies

$$\dot{g} = \partial_t g + \nabla_{\mathbf{z}} g \cdot \mathbf{f}^- \neq 0. \quad (6.4)$$

Near a transverse crossing, a small change in the residual gives a proportionally small change in event time. The first-order sensitivity is finite.

At a grazing event, $g = 0$ and $\dot{g} = 0$. Consider a point moving horizontally at height R past a circle:

$$g(t) = \sqrt{v^2 t^2 + R^2} - R. \quad (6.5)$$

It touches at $t = 0$ but g stays nonnegative on both sides. A detector that searches only for a sign change can miss it completely. The second derivative is positive at the touch, so the local trajectory returns to separation.

There are also sustained contacts: $g = 0$ for an interval while the constraints determine motion. An algorithm that treats every sample of that interval as a new impact will generate fictitious impulses. Event detection must therefore distinguish first contact, maintained contact and release.

6.4 Hysteresis and finite dwell time

Suppose a scalar input s drives a two-state model. With thresholds $s_{\text{low}} < s_{\text{high}}$,

$$\Psi^+ = \begin{cases} 1, & \Psi^- = 0 \text{ and } s \geq s_{\text{high}}, \\ 0, & \Psi^- = 1 \text{ and } s \leq s_{\text{low}}, \\ \Psi^-, & \text{otherwise.} \end{cases} \quad (6.6)$$

The state depends on history within the interval between thresholds. That dependence is explicit and can be tested. It differs from adding arbitrary jitter and assuming that noise must improve a threshold decision.

A minimum dwell time adds a timer reset at each transition and forbids another transition until the timer reaches a specified interval. A refractory model can do something similar through an internal recovery variable. Neither should be assigned to a particular insect without observation, but each is a concrete alternative architecture to instantaneous, memoryless switching.

A bounded input-error argument is also possible. If a stationary true signal is observed with error at most η , a threshold separation larger than 2η prevents that fixed true value from being driven across both thresholds by error alone. This is a derived property of the model, not a guarantee against genuine rapid changes in the signal.

6.5 A contact can change the future without changing position

An event reset may change velocity while leaving position continuous. It may release a constraint while leaving every coordinate initially unchanged. It may preserve the visible shape but change pressure or an internal state. Thus the visual appearance at the hinge instant can be insufficient to distinguish the futures.

A release event is especially clear. Just before and just after a latch opens, the geometry may be almost identical. Before release, the latch supplies a reaction that prevents motion; after release, it does not. The state and constraint structure have changed even before a large displacement is visible.

For a droplet, a topological change is different again: a liquid neck disconnects. A discretised level-set representation can describe separated components after the event, but the physical instant is not defined merely by one voxel changing sign. The neck dynamics and resolution criterion determine how the numerical event corresponds to the physical one.

6.6 Event time is a sensitive output

Perturb a transverse event. Differentiating $g(\mathbf{z}(t_*), t_*) = 0$ gives, for a small state perturbation $\delta\mathbf{z}^-$ evaluated at the nominal event time,

$$\delta t_* = -\frac{\nabla_{\mathbf{z}} g \cdot \delta\mathbf{z}^-}{\partial_t g + \nabla_{\mathbf{z}} g \cdot \mathbf{f}^-}. \quad (6.7)$$

Parameter perturbations add corresponding terms in the numerator. The denominator is the rate at which the trajectory crosses the event surface. Near grazing it becomes small, so small geometric or state errors can produce large timing differences.

This is why a clear-looking border can still produce an uncertain event time. Precision in the surface position and precision in temporal prediction are related through the crossing rate. A slow near-tangent encounter may be harder to time than a faster direct approach even when both surfaces are reconstructed to the same spatial accuracy.

6.7 The saltation matrix: sensitivity across the hinge

Across an event, perturbations change because of the reset and because perturbed trajectories reach the event at different times. For a single smooth transverse event, the first-order update is

$$\mathbf{S} = D_{\mathbf{z}}\mathcal{R} + \frac{(\mathbf{f}^+ - D_{\mathbf{z}}\mathcal{R}\mathbf{f}^- - \partial_t\mathcal{R})\nabla_{\mathbf{z}}g^\top}{\partial_t g + \nabla_{\mathbf{z}}g^\top\mathbf{f}^-}. \quad (6.8)$$

Here gradients are column vectors, all terms are evaluated at the nominal transition, and $\mathbf{f}^+$ is evaluated at the reset state. Then $\delta\mathbf{z}^+ = \mathbf{S}\delta\mathbf{z}^-$ to first order. The saltation term accounts for the timing contribution missing from the reset Jacobian alone. [6]

This is a direct technical expansion of the “hinge” idea. It tells us how differences before an event become differences afterward. A mechanism can suppress some perturbations, amplify others or redirect them into new coordinates. The result depends on the event geometry and both modes’ dynamics.

If a covariance $\mathbf{P}^-$ describes small local state uncertainty, an approximation is $\mathbf{P}^+ \approx \mathbf{S}\mathbf{P}^-\mathbf{S}^\top$, with additional terms for uncertain resets or other noise. This approximation is not a substitute for a multimodal distribution when some trajectories have transitioned and others have not. Simultaneous events also need an explicitly coupled or sequence-aware treatment.

6.8 Bifurcation is a separate question

A threshold crossing occurs along a trajectory under a defined condition. A bifurcation concerns a qualitative change in the structure of a dynamical system as a parameter varies. One does not imply the other.

For example, the scalar family $\dot{x} = \mu - x^2$ has two equilibria for $\mu > 0$, one repeated equilibrium at $\mu = 0$ and no real equilibrium for $\mu < 0$. That change is a saddle-node bifurcation. A binary output $\Psi = \mathbf{1}\{x > 0\}$ might change during an ordinary trajectory even when no bifurcation occurs. This small explicit model separates two concepts the source uses together.

For a periodically operating hinge, stability can be investigated through a return map over a full cycle, including both continuous flow sensitivities and event updates. A stable cycle, an unstable cycle and a single switching event are different mathematical objects. The framework becomes more useful when it can name each one accurately.

Space–time, margins and measured uncertainty

7.1 A moving boundary is a world tube

For a time-dependent body, collect its boundary through time:

$$\mathcal{W} = \{(\mathbf{x}, t) : d(\mathbf{x}, t) = 0\}. \quad (7.1)$$

This is the boundary’s history in space–time. A trajectory intersects that history when its position reaches the boundary at the same time. A static drawing of all positions swept by the body loses that timing information: two paths can occupy the same place at different times and never contact.

For one spatial dimension, a translating border $x_b(t) = x_0 + ut$ and a point $x_p(t) = x_1 + vt$ intersect when $x_p(t_*) = x_b(t_*)$. Provided $u \neq v$,

$$t_* = (x_0 - x_1)/(v - u).$$

The simple expression already captures the source’s temporal–spatial ambition more precisely than an unspecified four-dimensional rotation. It states which trajectories meet and the conditions under which they meet.

7.2 Spatial SDF versus a distance in space–time

The function $d(\mathbf{x}, t)$ is a spatial SDF at each instant. It is not automatically a signed distance to $\mathcal{W}$ in four-dimensional Euclidean space. Metres and seconds cannot be added in a norm without defining a conversion scale.

One possible analytical metric is

$$ds_*^2 = \|d\mathbf{x}\|^2 + v_*^2 dt^2, \quad (7.2)$$

where v_* is a declared speed scale. A distance in these coordinates measures nearness in a combined spatial–temporal representation. Its value depends on v_* . It is not automatically travel time, causal reachability or a physical spacetime interval.

For this reason, the basic hinge engine keeps spatial clearance and time-to-event as separate outputs. A combined metric is introduced only for a specified analysis task, such as matching two event histories or clustering trajectories after nondimensionalisation.

7.3 Distance, residual and probability are different margins

A geometric clearance margin has units of length. A pressure margin $p - p_c$ has units of pressure. An energy margin $E - E_c$ has units of energy. A probability that an event occurs before a deadline has no units. Each can be useful, but none becomes a spatial SDF simply because its sign separates two cases.

For a mixed state, define scaled coordinates

$$\tilde{z}_i = (z_i - z_{i,\text{ref}})/s_i \quad (7.3)$$

with positive scales s_i . A Euclidean distance in $\tilde{\mathbf{z}}$ is then dimensionless. Changing the scales changes the metric and the meaning of nearness. A more general positive-definite metric can weight correlated directions, but the choice remains part of the model.

Thus an “identity distance”, “cognitive distance” or “distance from a decision” requires explicit features and a metric. Those quantities do not inherit metres from the geometry of an insect or a rendered body. A measurable task variable, such as cursor distance to a target or elapsed time to a response, is a better starting point than an undefined global distance assigned to a person.

7.4 Spatial uncertainty becomes timing uncertainty

For a transverse encounter with approximately known normal approach speed $v_n > 0$, a first-order estimate is

$$\sigma_t \approx \frac{\sigma_g}{v_n}. \quad (7.4)$$

The benchmark in Chapter 11 uses $\sigma_g = 20 \mu\text{m}$ and $v_n \approx 0.285657 \text{ m/s}$, giving $\sigma_t \approx 70.0 \mu\text{s}$. The equation is a local propagation of an assumed gap uncertainty, not a new animal measurement.

With uncertain state covariance $\mathbf{P}$, a linearised residual variance is

$$\sigma_g^2 \approx \nabla g^\top \mathbf{P} \nabla g + \sigma_{\text{geom}}^2,$$

provided the added geometric error is appropriately modelled and independent of the state error. Correlated errors require their cross terms. Calibration drift shared by two cameras, for example, should not be counted as two independent noise sources.

A first-order Gaussian approximation may be poor near grazing, under multiple possible contacts or when event order is uncertain. In those cases, a distribution of simulated event times or explicit interval bounds can convey the model’s output more accurately than a single symmetric error bar.

7.5 Sample timing and the delta–delta calculation

For samples $x_k = x(t_k)$ at constant interval h , the centred differences are

$$\dot{x}_k \approx \frac{x_{k+1} - x_{k-1}}{2h}, \quad (7.5)$$

$$\ddot{x}_k \approx \frac{x_{k+1} - 2x_k + x_{k-1}}{h^2}. \quad (7.6)$$

A second derivative of position is acceleration. A third derivative is jerk. The same distinction applies to angular variables. The source’s “delta–delta” vocabulary becomes computationally meaningful when the measured variable, interval and derivative order are supplied. [U2, pp. 7–8].

If independent position errors have variance σ_x^2 , the error variance of the second-difference acceleration estimate is $6\sigma_x^2/h^4$. That follows by summing the squared coefficients 1, -2 , 1. A smaller sampling interval can therefore magnify derivative noise when spatial accuracy is unchanged. High frame rate and high derivative accuracy are not identical achievements.

Fit or filter choices also affect latency. A centred derivative uses a future sample and is unsuitable as an instantaneous causal estimate without delay. A real-time implementation must state whether it uses one-sided differences, a dynamical state estimator or a delayed output.

7.6 Reachability is more informative than closeness

A target can be close but unreachable because a joint limit blocks the motion. Another target can be farther away but reachable with a permitted trajectory. A short spatial distance therefore does not determine whether a mechanism can reach a boundary in time.

Let $\mathcal{U}$ be the admissible inputs and $\mathcal{G}$ the target event set. Define

$$\mathcal{R}_T = \{\mathbf{z}_0 : \exists \mathbf{u}(\cdot) \in \mathcal{U} \text{ such that the trajectory reaches } \mathcal{G} \text{ by } T\}.$$

This is a reachable-set definition. It requires dynamics, input limits and a time horizon in addition to geometry. Its boundary is a different type of border: it separates possible from impossible outcomes within the stated model and constraints.

A first-arrival field for a front moving with prescribed positive speed $c(\mathbf{x})$ can satisfy $\|\nabla T\| = 1/c(\mathbf{x})$. Fast-marching methods address monotonically advancing fronts of this kind. The field T has units of time; it is not a unit-gradient spatial SDF unless transformed under a special constant-speed setting. [7]

Operational consequence. Spatial location, temporal coincidence, state margin and dynamic reachability answer different questions. A complete hinge description reports the one required by the task rather than calling all of them distance.

Physical hinge mechanisms

The following eleven dossiers give the framework material content. Each begins with an observed mechanism from a primary study and then develops a specific modelling construction. The construction is not presented as a measurement of that animal. Its purpose is to identify the geometry, coupling, event and experiment needed for a useful explanation.

8.1 Intermeshing gears: *Issus* nymphs

Observed mechanism. Burrows and Sutton documented intermeshing gears on the hind legs of *Issus* nymphs. The teeth engage during propulsion and synchronise the leg movements. This is actual geared anatomy in a specified developmental stage, rather than a generic description of every insect joint. [8]

Model developed here. Let the two gear-bearing structures have local tooth surfaces with fields d_L^0 and d_R^0 . Their world-space surfaces are obtained through their measured poses, not by rotating an abstract label:

$$d_i(\mathbf{x}, \phi_i) = d_i^0 [R_i(\phi_i)^T (\mathbf{x} - \mathbf{c}_i)], \quad i \in \{L, R\}.$$

A candidate tooth pair has its own gap $g_j(\phi_L, \phi_R)$. While that pair remains in compressive contact, $g_j = 0$ constrains the angular velocities through

$$\frac{\partial g_j}{\partial \phi_L} \dot{\phi}_L + \frac{\partial g_j}{\partial \phi_R} \dot{\phi}_R = 0.$$

For ideal external circular gears, this reduces to $r_L \dot{\phi}_L + r_R \dot{\phi}_R = 0$ under compatible angle conventions. Equal pitch radii produce equal angular-speed magnitudes. Real tooth profiles, mounting geometry and compliance determine how closely this idealisation applies.

The consequential hinge is a *sequence of active contacts*. One pair of teeth transmits load; another takes over. Backlash gives an interval in which the geometric proximity does not yet impose the ideal rolling constraint. A model that merely commands two equal angles misses this distinction: command synchronisation and mechanical synchronisation are different mechanisms.

The direct investigative task is to recover both angular trajectories and the contact geometry. Compare the measured relative timing with a model that has tooth contact and with one that only has independent actuators. The discriminating object is the coupling, not whether the output can be written as two binary labels.

8.2 Trap-jaw ants: the same rotation, different external contact

Observed mechanism. In *Odontomachus bauri*, rapid mandible motion serves both prey-related functions and body propulsion when the jaws strike an external surface. Patek and colleagues connected the ballistic movement to the jaw apparatus and its latch-mediated operation. [9]

Model developed here. Represent the closing jaw by ϕ , an elastic coordinate by s , and the body pose by $\mathbf{q}_b$. The stored energy may be approximated locally as

$$U_s(s) = \frac{1}{2} k_s (s - s_0)^2.$$

A latch imposes a constraint while engaged. Release changes the admissible velocity or acceleration; the jaw then moves under the available elastic and muscular torques. The release event is separate from the later impact event.

There are at least two possible external gaps: g_p for a prey surface and g_s for a substrate. The active contact determines the Jacobian that enters the impact equation. A substrate contact can transfer an impulse to the body; a free prey object can acquire its own velocity. Equal jaw speed does not imply equal body motion, because the mass distribution and external constraints differ.

This is a concrete interpretation of the user's *pinion* vocabulary: the relevant object is the mechanism that constrains, releases and transmits motion. It need not be a literal toothed gear. The event record should distinguish latch disengagement, jaw acceleration, target contact and subsequent body or prey motion. Combining them into a single "strike bit" would discard the causal ordering that makes the mechanism informative.

8.3 Bombardier beetles: pressure feedback moves the boundary

Observed mechanism. Anatomical work and synchrotron X-ray imaging in living bombardier beetles linked spray pulsation to structures between the reactant reservoir and reaction chamber. The study supports mechanical feedback in which reaction-driven pressure displaces compliant structures that regulate renewed inflow. [10]

Model developed here. Let p be chamber pressure and q_v a valve-opening coordinate. The valve has a geometric gap $g_v(q_v)$; the chamber has a compliant volume $V(p, q_v)$. A reduced volume balance can be written

$$C_p \dot{p} + V_q \dot{q}_v = Q_{\text{in}}(p, q_v) - Q_{\text{out}}(p, q_v) + Q_{\text{rxn}},$$

where $C_p = \partial V / \partial p$ has units of volume per pressure, $V_q = \partial V / \partial q_v$, and Q_{rxn} represents an appropriately derived volume-generation term. Temperature, reaction progress and compressibility would be additional states in a more detailed model.

The valve dynamics can be represented by an effective mass, damping and restoring law:

$$M_v \ddot{q}_v + c_v \dot{q}_v + k_v(q_v - q_{v0}) = A_v \Delta p + F_{\text{contact}}.$$

Pressure is now connected to displacement through area and material response. The passage geometry changes the flow, and the flow changes the pressure. A pulsed discharge can emerge from this coupled loop without a separate digital command for every pulse.

The physical border is not just the outlet. It includes the valve's contact surface, the deforming chamber wall and the liquid–vapour interface. Those borders have different equations. An SDF can locate each interface; it cannot replace the chemistry, gas generation or material compliance that moves it.

8.4 Sharpshooter excretion: a deformable droplet on a moving stylus

Observed mechanism. Challita and colleagues measured sharpshooter droplet ejection and identified superpropulsion: droplets could move faster than the launching stylus. Their analysis connects the launch to liquid deformation and the temporal relationship between the droplet and actuator. [11]

Model developed here. Use $d_\ell(\mathbf{x}, t)$ for the liquid region and $d_s(\mathbf{x}, \phi)$ for the stylus. The droplet's boundary is a free surface, not a rigidly rotated sphere. Its surface energy is

$$E_\gamma = \gamma A,$$

with surface tension γ and interfacial area A . For a static interface with a suitable curvature convention, the pressure jump is $\Delta p = \gamma\kappa$. Shape change therefore changes both local pressure and stored surface energy.

There are three distinct events: the actuator begins its stroke, the connected liquid shape reaches a necking or release condition, and the detached droplet enters free flight. The zero set can deform continuously before pinch-off while its connectivity changes at detachment. Assigning an instantaneous reset to a coarse droplet model is a numerical representation of that change, not a claim that the fluid velocity is everywhere discontinuous.

An inertial–capillary time scale follows dimensionally as

$$t_\gamma = \sqrt{\frac{\rho R^3}{\gamma}}.$$

The ratio of stroke duration to t_γ is a more informative timing parameter than the word “fast.” A rigid projectile and a deformable droplet can respond differently to the same actuator trajectory. The model should track actuator work, droplet translation, internal deformation, residual attachment and dissipative loss separately.

8.5 Woodpeckers: contact as an impulse-transmission problem

Observed mechanism. Van Wassenbergh and colleagues used high-speed recordings and modelling to examine woodpecker pecking. Their results support a head acting as a stiff hammer with limited cranial shock absorption during the measured strikes, rather than a broadly cushioning skull. [12]

Model developed here. Let the beak tip have gap g to a substrate reconstructed by an SDF. The contact normal controls the direction of the normal impulse, while the lever arm to the head’s mass distribution contributes torque. A small contact offset can therefore alter rotational response without substantially changing the initial tip speed.

A local impact model must include the substrate’s deformation as well as the head’s effective mass. For example, a short compliant contact can use an overlap variable $\delta = -g$ and a compressive force $F_n(\delta, \dot{\delta})$ chosen for the material pair. The measured tip deceleration alone is not a measurement of how much energy every tissue absorbed.

This dossier shows why *transmission* belongs alongside capture and release in the framework. A boundary can transmit a large load while staying intact. The relevant event may be contact onset, peak force, material indentation or contact termination. Those moments should not all be named “impact” in a data log without further distinction.

A useful experiment would align tip position, head pose and substrate displacement to a common clock. Compare predictions for the onset time, impulse and rebound trajectory. Agreement on the contact time but disagreement on rebound identifies a dynamics problem rather than necessarily a geometric problem.

8.6 Penguin droppings: discharge, then trajectory

Observed mechanism. Meyer-Rochow and Gal estimated the pressures involved in penguin defaecation from geometry, fluid properties and observed discharge. Their pressure values were calculations from a model, not direct internal pressure recordings. [13]

Model developed here. Define an outlet surface Γ_o with outward normal $\mathbf{n}$. The mass flux relative to a moving outlet is

$$\dot{m}_{\text{out}} = \int_{\Gamma_o} \rho(\mathbf{v} - \mathbf{v}_o) \cdot \mathbf{n} \, dA.$$

This is the correct kind of boundary accounting: matter crosses a specified surface. It does not imply that the material has disappeared from the larger environment.

For an ideal Newtonian liquid in a straight circular tube with steady, fully developed laminar flow, the pressure drop is

$$\Delta p = \frac{8\mu\ell Q}{\pi a^4}.$$

Here the assumptions belong to the ideal model, not automatically to the complete biological tract or every consistency of ejecta. Yield stress, changing geometry, transient acceleration and a particulate mixture can require a different constitutive law.

Once an ejected parcel no longer exchanges substantial force with the outlet, a trajectory model takes over. Under a simple no-drag approximation, gravity sets the subsequent curvature of the path. With air resistance or breakup, parcel size and deformation become relevant states. Thus “expulsion” is not a single equation: it is a transition from pressure-driven internal flow to external motion, possibly with fragmentation in between.

The hinge record should identify which phase supplied each inferred quantity. Inferring launch speed from range is not the same operation as measuring chamber pressure. Both can be useful; their relationship must be calculated rather than assumed.

8.7 Duck gut passage: an exit is not a terminal state

Observed mechanism. Controlled feeding experiments recovered a small fraction of viable cyprinid fish eggs after passage through ducks, and some recovered eggs subsequently hatched. The experiment establishes possible survival through that route; it does not by itself measure successful establishment in a new natural waterbody. [14]

Model developed here. Geometric ingress and egress are separate events from loss of viability. Let the egg position be $\mathbf{x}_e(t)$, its viability-related state be $a(t)$, and g_o denote passage through an outlet surface. An exit occurs when the trajectory crosses $g_o = 0$ in the specified direction. A damage condition can occur earlier or later.

One possible reduced survival model introduces a nonnegative hazard $k(t)$ and

$$P(\text{survival to } t) = \exp\left[-\int_0^t k(s) ds\right].$$

This is a proposed statistical representation. The hazard would need to be identified from exposure, residence time and observed outcomes. It is not supplied by the distance field.

A complete dispersal sequence contains conditional stages. Its probability can be factorised as

$$\begin{aligned} P(\text{establishment}) &= P(\text{ingestion}) P(\text{viable exit} \mid \text{ingestion}) \\ &\quad \times P(\text{suitable deposition} \mid \text{viable exit, ingestion}) \\ &\quad \times P(\text{establishment} \mid \text{preceding stages}). \end{aligned}$$

This is the chain rule, not an independence assumption. It shows how a binary end result can depend on multiple continuous trajectories and conditional events. An excretory boundary can be a transport link rather than the final endpoint of the transported object.

8.8 Seabird guano: local deposition, regional atmospheric consequences

Observed mechanism. Croft and colleagues combined atmospheric observations and modelling to study ammonia associated with Arctic seabird colonies. Their results connect colony-derived ammonia with particle formation and a modelled cloud-albedo radiative effect in the studied setting. [15]

Model developed here. The deposited material starts a sequence of inventories and transfers. Nitrogen in a ground deposit is not the same state variable as gas-phase ammonia, aerosol number concentration or a cloud optical property. A compartment balance for gas-phase concentration c might begin with

$$\frac{\partial c}{\partial t} + \nabla \cdot (c\mathbf{u}) = \nabla \cdot (D\nabla c) + S_{\text{emit}} - L_{\text{chem}} - L_{\text{dep}}.$$

All terms must have matching concentration-per-time units. Emission depends on the deposit and its environment; aerosol formation requires additional chemistry and particle dynamics. A threshold in one of those states is not a centimetre-scale SDF.

Several borders matter: the material–air interface, the boundary of an atmospheric control volume, and a chosen concentration contour. The contour $c = c_*$ is an isosurface but is generally not a signed-distance field. Its gradient magnitude depends on the concentration distribution.

This is a strong example of the framework extending across scales without making those scales identical. The initial dropping is spatially local. Its components may be transported and transformed. The resulting effects depend on humidity, temperature, mixing and chemical partners. The event graph has branching and delays, not one universal switch repeated at larger size.

8.9 Fungal occupation of ant tissues: location before control claims

Observed mechanism. Three-dimensional microscopy and image analysis revealed extensive fungal networks in behaviourally manipulated ants, including muscle-associated structures, while the brain was not invaded in the examined material. These observations locate fungal structures; they do not identify a universal digital control bit or provide a complete causal account of every behavioural change. [16]

Model developed here. Let Ω_h denote host tissue compartments and $\Omega_f(t)$ the segmented fungal material. Signed distance can quantify proximity between the fungal surface and a muscle boundary, while overlap measures can quantify occupancy. A surface-distance distribution is more informative than a single image labelled “takeover.”

To connect location to function, introduce measurable tissue variables: local deformation, contractile force, damage, chemical concentration or neural activity. A hypothetical coupling law such as

$$F_{\text{muscle}} = F_0(\ell, \dot{\ell}, a) [1 - D(t)]$$

requires independent definitions of activation a and damage D . It is a testable reduced model, not a conclusion obtained simply by seeing fungal material nearby.

The hinge may lie in a change of attachment, tissue integrity or force generation. Establishing which one requires time-resolved evidence. Spatial co-location supports a candidate mechanism; causal identification requires determining how a change in one variable changes another. This keeps the investigative focus on the physical interface without inventing the unobserved internal software of an organism.

8.10 Plant volatiles: a chemical boundary, not a radio command

Observed mechanism. Turlings, Tumlinson and Lewis found that caterpillar feeding and oral secretions elicited volatile release in maize seedlings, and that female parasitoid wasps could learn to use associated

plant odours to locate hosts. The experiment connects elicitation, chemical emission and behaviour under its tested conditions. [17]

Model developed here. The source’s wordplay “pherrormones” can name a project interest in chemically mediated interactions. The physical variables must still be concentrations, transport coefficients, receptor states and measured responses. [U1, pp. 20–22].

Let $c(\mathbf{x}, t)$ be the concentration of one identified compound, transported by advection and diffusion. A reduced receptor-occupancy model at a receiver location $\mathbf{x}_r(t)$ is

$$\dot{b} = k_{\text{on}}c(\mathbf{x}_r, t)(1 - b) - k_{\text{off}}b, \quad 0 \leq b \leq 1.$$

This model contains memory: identical current concentrations can accompany different occupancies because the recent exposure histories differ. A response event could use $g_b = b - b_*$, together with a direction and a behavioural context. Neither b nor g_b is a spatial distance.

A moving receiver also crosses concentration contours. Its experienced rate of change is

$$\frac{dc(\mathbf{x}_r, t)}{dt} = \partial_t c + \dot{\mathbf{x}}_r \cdot \nabla c.$$

The same mathematical chain rule that describes a moving point approaching a surface now describes sampling a chemical field. The shared operation is calculus; the fields retain different physical units and mechanisms. It does not turn odour molecules into radio waves.

8.11 Rainbow optics: the material surface and the ray-map fold

Physical basis. Refraction and dispersion depend on the electromagnetic response of matter. A material’s refractive index can vary with frequency; this should not be described as coloured particles experiencing different amounts of mechanical friction. Feynman’s treatment develops refractive index through the response of charges to electromagnetic waves. [18]

Model developed here. A spherical water interface can be represented by $d(\mathbf{x}) = \|\mathbf{x}\| - R$. A ray reaches the interface where $d = 0$. The field gradient supplies the surface normal. Refraction then follows an optical law; the existence of the surface alone does not determine the new direction.

For an ideal spherical drop, a ray with one internal reflection has angular radius from the antisolar direction

$$\alpha(i, \lambda) = 4r(i, \lambda) - 2i, \quad \sin i = n(\lambda) \sin r.$$

Holding wavelength fixed gives

$$\frac{\partial \alpha}{\partial i} = \frac{4 \cos i}{n \cos r} - 2.$$

The stationary-angle condition is $2 \cos i = n \cos r$. Combining it with Snell’s law gives $\cos^2 i = (n^2 - 1)/3$ in the ordinary primary-rainbow configuration. Wavelength dependence in n shifts the stationary angle.

There are now two clearly different borders. The first is a *material interface*, $d = 0$. The second is a *fold of the ray mapping*, where the derivative of outgoing angle with respect to the incident-ray parameter vanishes. The latter concentrates geometrical ray density; finite wave effects and the source’s angular extent regularise the ideal singular description.

This is a precise way to retain the user’s spectrum–bifurcation interest. A continuous spectrum, an optical fold and a discrete sensor threshold can appear in the same chain without becoming the same mathematical object. The rainbow’s optical mechanism is fully physical. What a person chooses to symbolise with it is an additional interpretive layer, not a term in the ray equation.

Cognition, creative systems and engineered hinges

9.1 The object of the cognitive investigation

The job description includes cognition because a mechanism-centred practice can investigate how measurements become estimates, how estimates affect actions, and how interfaces alter the available feedback. The useful unit is a specified sensing–inference–action task. It is not a geometrical claim about the whole identity of a person.

A task model might contain an external state $\mathbf{x}$, a measurement $\mathbf{y}$, an internal estimate $\hat{\mathbf{x}}$ and an action $\mathbf{u}$:

$$\mathbf{y} = h(\mathbf{x}) + \boldsymbol{\eta}, \quad \dot{\hat{\mathbf{x}}} = f_{\text{est}}(\hat{\mathbf{x}}, \mathbf{y}), \quad \mathbf{u} = \pi(\hat{\mathbf{x}}, m).$$

These equations define an engineering abstraction. They do not establish that a nervous system literally runs the selected software or uses only the listed variables. Their value is that an experiment can independently record stimuli, responses and the model's predictions.

Within this abstraction, a hinge can be a change of selected action, a contact in the external world, or a revision of an internal estimate. Recording all three with the same label would make it impossible to determine whether a failure began in sensing, estimation or action.

9.2 A local decision boundary can produce a bit

Model developed here. Consider a scalar evidence state a with leakage and time-varying input:

$$\dot{a} = -\lambda a + u(t).$$

A decision rule can emit an event when a first reaches a boundary a_* from below. Add a reset $a^+ = a_r$ or switch the model into another mode. This is a fully defined continuous-state/discrete-event construction. It gives precise content to a local Ψ bit without reducing the entire system to that bit.

Two trajectories can produce the same output $\Psi = 1$ after very different input histories. Conversely, the same current input can produce different imminent decisions if a differs. The state is what preserves the history that a one-bit output has discarded.

The signed margin $g_a = a - a_*$ is useful, but its units are evidence units defined by the model. Calling it a distance in metres would be false. To compare two task dimensions, first establish their scaling and metric. For a multidimensional feature vector $\mathbf{a}$, a boundary $F(\mathbf{a}) = 0$ can be turned into a distance only after the feature-space geometry is specified.

The practical experiment is to fit parameters on one set of tasks and evaluate event times on separate tasks. A visually compelling threshold animation is not a substitute for testing the timing and error pattern.

9.3 Extended reality: a visible boundary with a measurable delay

Model developed here. In an extended-reality scene, a tracked controller or hand proxy approaches a virtual object. Let d_V be the object's field and $\mathbf{p}(t)$ the tracked proxy position. A contact event can be

defined by

$$g(t) = d_V(\mathbf{p}(t)) - r_p = 0,$$

where r_p is a spherical proxy radius. The radius is a modelling choice. A fingertip mesh, a stylus or a whole articulated hand needs a different geometry.

The scene has several clocks: sensor exposure, pose estimation, simulation update, rendering and display. For a proxy approaching at approximately constant normal speed v_n , a net latency τ produces a first-order apparent normal displacement of roughly $v_n\tau$. This follows from motion during the delay. It should not be mistaken for an error in the object's static SDF.

The event can trigger a visual deformation, a sound or a haptic response. Each output has its own latency and dynamics. A good mechanism studio therefore exposes the timing chain. It allows the user to compare the predicted contact, the sampled contact, the rendered contact and the actuator response on one timeline.

The creative opportunity is not merely to colour the object. It is to make an otherwise hidden relation inspectable: how a normal changes along a surface, why a shallow grazing contact is timing-sensitive, or why a small contact offset changes torque. A scene becomes an instrument when the displayed quantities can be traced back to defined variables.

9.4 Bugfinding as locating a broken explanatory link

A mechanism-centred bugfinder asks where the observed sequence first departs from the specified sequence. The diagnostic hierarchy is practical.

A **geometry discrepancy** changes the location or normal of the surface. A **frame discrepancy** places correct local geometry in the wrong world pose. A **dynamics discrepancy** produces the wrong acceleration or impulse despite correct geometry. A **state discrepancy** omits damage, memory or a mode needed to predict the next event. A **timing discrepancy** evaluates the right condition at the wrong time. A **representation discrepancy** changes the boundary while pretending only its display has changed.

This vocabulary connects the professional identity to a repeatable technical act. An insect observation, a robot collision, a fluid detachment and an interface error can all motivate the same question: which relation in the model is responsible for the mismatch? The answers remain system-specific.

One productive residual is the event-time error

$$r_{t,k} = t_{k,\text{observed}} - t_{k,\text{predicted}}.$$

Another is the geometric residual evaluated at observed contact,

$$r_{g,k} = g(\mathbf{z}_{k,\text{observed}}, t_{k,\text{observed}}).$$

A systematic nonzero r_g points toward shape, registration or event-definition problems. A near-zero r_g with an incorrect post-contact trajectory points toward the force or reset law. This is a diagnostic inference to be tested, not an automatic guarantee about the cause.

9.5 Representation and the executive vision

The source contrasts symbols with the physical mechanisms they represent. Here this becomes a technical principle: *the display must remain traceable to the mechanism*. [U1, pp. 16–20, 37–41].

A smooth image can display a discontinuous controller; a binary icon can represent a continuous process. The proposed practice therefore preserves the chain from observation to geometric model, from model to event calculation, and from calculation to rendered or engineered output. Changing the presentation must not silently change the specified physical model.

A computational specification for the hinge engine

10.1 Separate geometry, dynamics and event services

The proposed implementation has three primary services. The geometry service evaluates a field, a normal and a representation error. The dynamics service advances the state in its current mode. The event service determines whether an eligible transition has occurred, locates its time and applies the defined reset. Keeping these services separate makes each one testable.

For geometry, a minimal interface is

$$\mathcal{G}(\mathbf{x}, t) \mapsto (d, \mathbf{n}, \varepsilon_d, \text{frame}, \text{validity}).$$

The validity field matters. A normal can be non-unique at a medial axis or geometric corner. A truncated field can be saturated. A reconstructed surface can be outside the observed volume. Returning an ordinary-looking number without its validity state can produce an apparently precise but invalid contact calculation.

For events, the corresponding interface is

$$\mathcal{E}_i(\mathbf{z}, t, m) \mapsto (g_i, \dot{g}_i, \text{direction}, \text{eligibility}, \mathcal{R}_i).$$

An event condition may depend on more than geometry. A contact-release condition may involve a force; a fatigue condition may involve accumulated damage; a sampled decision may be eligible only when new data arrive. The event service should not assume that every condition is an SDF.

10.2 Choose a representation for a specified purpose

Analytic geometry. Spheres, planes, capsules and boxes provide reference cases with known distances. They are the correct starting point for verifying normals, transforms, contact roots and numerical tolerances. Their simplicity makes errors visible.

Triangle geometry. A closed, consistently oriented surface can support nearest-surface queries and an inside/outside classification. Nearest distance alone is unsigned; the sign requires a separate interior convention. A thin open sheet does not automatically define a solid interior. Self-intersections and missing faces require explicit handling rather than an invented sign.

Sampled fields. A voxel field stores values at discrete locations. Interpolation yields a convenient query function, but its gradient and zero crossing can differ from the continuous surface used to create it. Grid spacing is a numerical scale, not a proof that smaller geometric features were measured.

Truncated and Euclidean fields. Voxblox describes an approach that incrementally constructs Euclidean signed distance information from a truncated signed distance representation for planning. This is a useful example of choosing one representation for integration and another for geometric queries. The source does not make every truncated sensor residual an exact Euclidean distance. [19]

Learned implicit fields. DeepSDF learns continuous shape representations through a signed-distance formulation. In a hinge engine, a learned field would still need testing for surface accuracy, gradients,

extrapolation and whether it supplies a conservative distance bound for the chosen operation. Learning a function named SDF does not make its every evaluation analytically exact. [20]

10.3 Keep the zero set and the distance property distinct

A transported level-set function can continue to locate the correct interface while losing unit gradient away from it. A reinitialisation step can seek a distance-like function without changing the intended interface. A common formulation introduces artificial iteration time τ :

$$\partial_\tau d + S(d_0)(\|\nabla d\| - 1) = 0,$$

where S is a suitable sign approximation based on the initial field. Here τ is a numerical variable, not physical time. Level-set methods for two-phase flow develop this separation between interface tracking and maintaining a useful field representation. [4]

In a numerical implementation, monitor both the Eikonal residual and surface displacement. Decreasing

$$E_{\text{eik}} = \left(\frac{1}{N} \sum_{j=1}^N (\|\nabla d(\mathbf{x}_j)\| - 1)^2 \right)^{1/2}$$

is not a success if the zero set has moved far enough to change the predicted first contact. The geometry service should report the maximum or distribution of that displacement in the region used by the experiment.

Smoothing has the same issue. A smooth minimum can improve numerical derivatives while altering both surface location and metric distance. An executive specification should therefore state whether a smooth field is a display surface, a contact surface or only a broad-phase proximity estimate. These roles have different error requirements.

10.4 Conservative stepping needs an actual bound

Distance-based ray stepping is useful because an exact exterior distance gives a region around the current point that contains no surface. Hart's sphere-tracing formulation exploits distance and related conservative bounds for implicit-surface intersection. [2]

For a moving mechanism, time stepping needs a motion bound as well as a distance. Suppose $g(t_0) > 0$ and the closing speed is bounded by $-\dot{g} \leq V_{\max}$ throughout the next interval. Then

$$g(t) \geq g(t_0) - V_{\max}(t - t_0).$$

Consequently no contact can occur before $g(t_0)/V_{\max}$ under those assumptions. With distance uncertainty ε_g , use a valid lower gap bound rather than the nominal value. The bound must include target motion, rotation and shape change where relevant.

A general implicit function F can provide a conservative distance estimate when a valid Lipschitz constant L is known: $|F(\mathbf{x})|/L$ is no greater than the distance to any zero of F in the domain over which the bound applies. This follows directly from

$$|F(\mathbf{x}) - F(\mathbf{y})| \leq L \|\mathbf{x} - \mathbf{y}\|, \quad F(\mathbf{y}) = 0.$$

An estimated local gradient is not automatically a valid global Lipschitz bound. Overstepping becomes possible when the bound is violated.

10.5 Locate the event before applying the reset

The basic execution sequence is:

1. Advance the continuous state provisionally in its current mode.
2. Evaluate every eligible event condition, including possible grazing or contact persistence where the model requires them.
3. Find the earliest relevant event time, using the trajectory interpolant and an appropriate root or minimum search.
4. Re-evaluate the state at that time, apply the selected reset or constraint change, and continue in the resulting mode.

A bisection routine only solves a bracket it has been given. It does not establish that the bracket contains the first event or detect a root that merely touches zero. A step can also contain two crossings whose endpoint signs agree. The collision-detection layer is therefore more than a sign comparison.

After impact, the new mode may be separation, sustained contact, sliding or sticking. Immediately testing the same zero-valued condition without a mode or direction rule can create numerical chatter. A physically derived contact mode, an appropriate complementarity method or a specified hysteretic event rule is preferable to adding an unexplained arbitrary delay.

Simultaneous contacts need a coupled treatment or a stated priority rule. A sequence of independent pairwise impulses can depend on the chosen processing order. That processing choice is part of the implementation, not an invisible property of nature.

10.6 Record an event as a reproducible object

An event log needs enough information to reconstruct the calculation. The following schema is a proposed minimum; the numbers illustrate structure rather than a measured animal.

```
{
  "event_id": "hinge-plane-001",
  "model_id": "planar-hinge-v1",
  "units": "SI",
  "frame": "world",
  "time_s": 0.004046011698,
  "event_type": "normal_contact_onset",
  "guard_definition": "a - L*cos(phi) - tip_radius",
  "crossing_direction": "positive_to_zero",
  "mode_before": "free_rotation",
  "mode_after": "impact_reset",
  "distance_representation": "analytic_plane",
  "geometry_status": "synthetic_benchmark",
  "reset_model": "single_frictionless_restitution",
  "parameter_set": "benchmark-11.1"
}
```

A production record should also retain the actual parameter values, software revision, root bracket, tolerance, normal, pre-event state, post-event state and measurement provenance. A visual label can then be traced to the calculation that created it.

10.7 The supplied computational companion

The standalone file `Hinge_Field_Companion_Tom_Klootwijk.py` implements exact sphere and box fields, a rigid-transform pullback, Rodrigues rotation, a planar hinge, a bracketed event solver, a hysteresis example and an analytic bouncing-ball verification. It uses the Python standard library and is written for Python 3.10 or later.

Run the benchmark report with:

```
python Hinge_Field_Companion_Tom_Klootwijk.py
```

Run the verification tests with:

```
python Hinge_Field_Companion_Tom_Klootwijk.py --test
```

The twelve tests cover field signs, a box corner, rotation-axis invariance, length preservation, rigid pullback, the hinge contact root, its derivative, an unbracketed grazing contact, hysteretic memory, a non-unique sphere normal, invalid inputs and saltation against a finite-difference trajectory derivative.

The implementation intentionally rejects invalid geometry such as a zero rotation axis or a nonpositive radius. The event solver requires a supplied sign-changing bracket. These are specifications of what the functions compute. They prevent an uncomputed quantity from being silently returned as a result.

Operational consequence. The engine should make its boundaries inspectable twice: once as physical or mathematical surfaces, and once as the conditions under which each numerical function is valid.

Worked calculations and verification results

All numbers in this chapter belong to synthetic analytic benchmarks. They demonstrate the equations and test the implementation. None is presented as a measurement from the user's photographs or a fitted parameter set for a particular animal.

11.1 A rounded tip rotating toward a plane

Consider a planar hinge at the origin. The tip centre has position

$$\mathbf{p}(\phi) = L(\cos \phi, \sin \phi),$$

and the target is the half-space $x \geq a$. Its outward normal on the near face is $\mathbf{n} = (-1, 0)$. A rounded tip of radius r_p has exterior gap

$$g(\phi) = a - L \cos \phi - r_p. \quad (11.1)$$

The angle starts at ϕ_0 and decreases at constant rate $\omega > 0$:

$$\phi(t) = \phi_0 - \omega t.$$

The synthetic parameters are $L = 4$ mm, $a = 3$ mm, $r_p = 0.2$ mm, $\phi_0 = 1.2$ rad and $\omega = 100$ rad s⁻¹.

11.1.1 Contact time

At contact,

$$\cos \phi_* = \frac{a - r_p}{L} = 0.7, \quad \phi_* = \arccos(0.7) = 0.7953988302 \text{ rad.}$$

On the selected decreasing-angle branch, the first contact occurs at

$$t_* = \frac{\phi_0 - \phi_*}{\omega} = 4.046011698 \text{ ms.}$$

The initial gap is 1.350568982 mm. Bisection on the supplied bracket returns 4.046011699 ms. The error relative to the analytic value is approximately 5.07×10^{-13} s in the recorded run.

11.1.2 Normal approach speed

The gap derivative is

$$\dot{g} = L \sin \phi \dot{\phi}.$$

At contact, the gap Jacobian is

$$J_g = L \sin \phi_* = 2.856571371 \text{ mm rad}^{-1},$$

and $\dot{g}_* = -0.2856571371 \text{ m s}^{-1}$. The closing-speed magnitude is therefore $0.2856571371 \text{ m s}^{-1}$.

This is the difference between angular speed and approach speed. The same 100 rad s^{-1} can generate a different normal approach speed when the contact point or orientation changes. Near a configuration where $J_g = 0$, the instantaneous normal speed vanishes even though the hinge rotates.

11.1.3 Normal force from a quasistatic torque balance

Suppose the actuator applies a closing torque of magnitude $2 \mu\text{N m}$ and the contact is quasistatic, frictionless and the only balancing load. With the stated sign convention, the contact force magnitude is

$$\lambda = \frac{|\tau|}{J_g} = 0.000700140042 \text{ N} = 0.700140042 \text{ mN}.$$

This is a static calculation. It should not be applied as the transient impact force of the incoming tip. Impact requires an impulse or a resolved compliant force history.

11.1.4 Impact with a specified effective inertia

Now use a fixed-pivot hinge inertia $I = 2 \times 10^{-11} \text{ kg m}^2$ and coefficient of normal restitution $e = 0.4$. Immediately before contact, $\dot{\phi}^- = -100 \text{ rad s}^{-1}$.

For this single degree of freedom and stationary target,

$$p_n = -\frac{(1+e)\dot{g}^-}{J_g^2/I} = 9.801960588 \times 10^{-7} \text{ N s}.$$

The corresponding post-impact angular speed is $\dot{\phi}^+ = 40 \text{ rad s}^{-1}$. The normal velocity has reversed according to the restitution rule. The kinetic energies are

$$E^- = \frac{1}{2}I(100)^2 = 1.0 \times 10^{-7} \text{ J}, \quad E^+ = 1.6 \times 10^{-8} \text{ J}.$$

The retained fraction is $e^2 = 0.16$ in this particular one-degree-of-freedom model. Changing the restitution or adding other degrees of freedom changes the result. No energy conservation follows from calling the motion a rotation.

11.1.5 Geometry uncertainty becomes timing uncertainty

For a gap standard deviation of $20 \mu\text{m}$ and negligible uncertainty in the closing speed, the first-order timing standard deviation is

$$\sigma_t \approx \frac{\sigma_g}{|\dot{g}_*|} = 70.0140042 \mu\text{s}.$$

This is over eight orders of magnitude larger than the recorded root solver's error relative to the analytic benchmark. More root-finding digits would not resolve a $20 \mu\text{m}$ geometry uncertainty. The arithmetic is precise conditional on the geometry; the event prediction remains sensitive to how well that geometry is known.

11.2 An approaching point and a moving, expanding sphere

Let a sphere have centre $c(t) = 0.05t$ metres along the x -axis and radius $R(t) = 0.010 + 0.01t$ metres. A point approaches along the same axis from $x_p(t) = 0.012 - 0.15t$ metres. Until contact, it remains on the positive side of the centre.

The exact exterior gap is

$$g(t) = x_p(t) - c(t) - R(t) = 0.002 - 0.21t.$$

Hence

$$t_* = 9.523809524 \text{ ms}.$$

The 0.21 m s^{-1} gap-closing rate contains three contributions: point motion, centre motion and radial expansion. A calculation using only point speed would predict 13.333 ms and would be late. A rigid-body target velocity would account for the centre motion but would still miss the expansion term.

The relevant moving-boundary derivative is therefore $\partial_t d + \nabla d \cdot \dot{\mathbf{x}}_p$, not merely the dot product of a stationary normal with the point velocity.

11.3 A grazing event that endpoint signs do not reveal

Consider a circle of radius R centred at the origin and a point on the tangent path $\mathbf{p}(t) = (vt, R)$. Its field value is

$$g(t) = \sqrt{v^2 t^2 + R^2} - R.$$

The point touches at $t = 0$, where $g = 0$ and $\dot{g} = 0$. On both sides, $g > 0$. With $R = 5 \text{ mm}$, $v = 0.2 \text{ m s}^{-1}$ and samples at $t = \pm 5 \text{ ms}$, both endpoint gaps are approximately $99.0195 \mu\text{m}$.

A sign-change detector sees no crossing. An algorithm that also examines the minimum gap or uses a valid continuous-collision procedure can locate the touch. The distinction is not numerical pedantry: a tangential touch may or may not trigger a physical effect depending on contact duration, adhesion and force. The geometric event has to be identified before those consequences can be modelled.

11.4 Capillary timing and a ballistic discharge

For a synthetic water-like drop with $R = 100 \mu\text{m}$, $\rho = 1000 \text{ kg m}^{-3}$ and $\gamma = 0.072 \text{ N m}^{-1}$, the inertial-capillary time scale is

$$t_\gamma = 117.8511302 \mu\text{s}.$$

This is a scale obtained from the stated material parameters and radius. An attached, viscous or nonspherical drop does not necessarily oscillate as an isolated inviscid sphere. The companion also reports the classical ideal free-drop quadrupole frequency using $\omega_2^2 = 8\gamma/(\rho R^3)$; that optional calculation retains the free-drop assumptions. The same frequency expression is used in the sharpshooter analysis. [11]

For a separate synthetic horizontal discharge from height $h = 0.20 \text{ m}$ with launch speed $u = 2.0 \text{ m s}^{-1}$, neglect drag and take $g_0 = 9.81 \text{ m s}^{-2}$. The flight time and range are

$$t_f = \sqrt{\frac{2h}{g_0}} = 0.2019275109 \text{ s}, \quad x_f = ut_f = 0.4038550219 \text{ m}.$$

The release model supplies the initial position and velocity. The external trajectory calculation then predicts where that parcel arrives. Neither calculation alone determines the pressure inside the animal that produced the discharge.

11.5 One threshold versus hysteresis

Use the following synthetic scalar input sequence:

$$-0.20, 0.02, -0.03, 0.04, -0.02, 0.20, 0.01, -0.02, 0.03, -0.20.$$

A memoryless rule $\Psi = \mathbf{1}[g > 0]$ gives eight state changes between consecutive samples. A Schmitt rule that switches on only above $+0.10$ and off only below -0.10 , starting off, gives two.

Sample	1	2	3	4	5	6	7	8	9	10
Hard threshold	0	1	0	1	0	1	1	0	1	0
Hysteretic state	0	0	0	0	0	1	1	1	1	0

This calculation supplies an exact example of the user’s “1-bit dither” interest. The output remains binary, but its response to jitter depends on whether the mechanism retains state. Added noise does not universally sharpen an event; here a memory rule suppresses repeated toggling caused by small sign changes.

11.6 Saltation tested against an entire trajectory

For a bouncing point mass, let the continuous state be $\mathbf{z} = (y, v)$, the flow be $\mathbf{f} = (v, -g_0)$ and the event surface be $y = 0$. At impact, use the reset

$$\mathcal{R}(y, v) = (y, -ev).$$

For an incoming velocity $v^- < 0$ and a stationary floor, the saltation matrix is

$$\mathbf{S} = \begin{bmatrix} -e & 0 \\ -\frac{(1+e)g_0}{v^-} & -e \end{bmatrix}.$$

The upper-left entry is not 1, even though the reset leaves y unchanged at the instant of impact. Perturbing initial height changes *when* the impact happens, which changes the trajectory at a fixed later observation time. Saltation includes that timing effect. [6]

Between impacts, the perturbation propagation over duration Δt is

$$\mathbf{F}(\Delta t) = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}.$$

For one isolated impact, the full fixed-time derivative is

$$\mathbf{D} = \mathbf{F}(T - t_*) \mathbf{S} \mathbf{F}(t_*).$$

The companion tests this matrix against finite differences of an analytic one-bounce trajectory with $y_0 = 0.20 \text{ m}$, $v_0 = -0.5 \text{ m s}^{-1}$, $T = 0.22 \text{ s}$ and $e = 0.6$. The verification passes at the implemented tolerance.

This is a stronger test than checking the impact velocity alone. It verifies that the event-time shift and the post-event flow are both included. It also supplies a concrete calculation for the “border of hinge events”: the boundary changes not only the nominal state but the sensitivity of the future trajectory.

11.7 Verification record

The supplied implementation completed twelve tests successfully. The test suite includes negative cases: it rejects a zero rotation axis, rejects a nonpositive radius and does not pretend that a grazing event has a sign-changing bracket. The benchmark output is reproduced from the same file that performs the calculations.

Executive program: from observation to invention

12.1 The executive mandate

Integrated executive vision

I build a practice for investigating mechanisms across living, cognitive and engineered systems. The organising problem is the consequential boundary: where contact begins, a constraint releases, a signal becomes actionable, material crosses an interface or a process changes its available future. I turn those boundaries into explicit models, experiments and technical instruments. The objective is an inspectable chain from observation to mechanism to working system.

This gives **Principal Transdisciplinary Technologist** a specific mandate. Technical leadership means owning the definition of the problem, the interfaces between disciplines and the conditions under which an output is accepted. **Computational Naturalist** gives that mandate its distinctive method: move between observed phenomena and computational structures, then return to the phenomenon to determine what the structure explains.

The hinge framework supplies a common technical language without requiring every project to use the same solver. A fluid interface, a jaw contact, a sensory decision and a rendering event need different equations. They can still share a disciplined event record, a common account of units and time, and an explicit relationship between the model and its observations.

12.2 Three coordinated workstreams

The interface observatory develops the source material: images, trajectories, surface reconstructions, timestamps and mechanism dossiers. Its primary output is an observation that another calculation can use. A useful record contains the species or object identification, developmental or operating state, imaging geometry, sampling interval, calibration and the precise event being described.

The hinge engine implements the model. Its outputs are field queries, transformations, contact conditions, state equations, event localisation and reproducible calculations. It exposes the difference between a geometric distance, a state margin and a probability. It also records whether a predicted event is well separated from alternative events or is sensitive to unresolved detail.

The mechanism studio builds the interactive or physical instrument: a simulation, an extended-reality scene, a demonstrator, an experimental controller or a design prototype. Its purpose is to make the mechanism usable. The displayed boundary should be tied to the same geometric and temporal definitions used by the engine, so changing the presentation does not silently change the model.

The streams exchange concrete objects. The observatory supplies calibrated data; the engine returns predictions and missing-variable requests; the studio makes the interaction inspectable and produces new observations. This is a workflow, not a claim that one software pipeline already explains every case in the atlas.

12.3 A sequence of acceptance decisions

The first acceptance decision is *definition*. State which event is being predicted and what counts as observing it. “Capture” needs a retention condition; “release” needs a lost connection or constraint; “failure” needs a specified change in function or material state. A visually dramatic frame is not by itself a complete event definition.

The second decision is *geometric adequacy*. Test field signs, distances, normals, frame registration and the position of the zero set. Measure whether geometric uncertainty is small enough for the required timing accuracy using the normal approach speed. A system can pass an image-quality check and still fail this contact-timing requirement.

The third decision is *dynamical adequacy*. Compare trajectories, impulses, force histories or transport rates. Correct contact location with incorrect rebound is a different failure from an incorrect surface. Identify which model term changes the prediction in the direction required by the observation.

The fourth decision is *event adequacy*. Test the predicted event sequence on data not used to set the parameters. Record missed events, false events, incorrect ordering and timing residuals. Include grazing and near-simultaneous cases rather than only well-separated crossings.

The fifth decision is *instrument adequacy*. Determine whether the prototype preserves the model’s timing, frames and units. The rendering can be smooth while the controller is late; the controller can be correct while the displayed object is displaced. Verify the complete chain that the user or experiment actually encounters.

12.4 What the practice delivers

A **mechanism dossier** contains the observed process, a diagram or geometry, state variables, coupling law, event definition and a discriminating test. A **computational model** contains executable equations, parameter provenance and regression tests. A **technical instrument** turns the model into an experimental or creative interaction. An **invention brief** identifies a transferable physical principle, the proposed engineered implementation and the experiments needed to determine whether the transfer works.

This separation makes the executive position actionable. A mechanism can inspire an invention without every detail of the organism being copied. The transferable principle might be a contact-controlled timing relation, a deformable launch interface or a pressure-feedback valve. The engineered material, scale and operating environment then determine whether the principle remains effective.

A result should be accepted because it predicts or demonstrates the chosen function, not because the terminology is elaborate. The professional distinction lies in joining the methods required to make that test possible.

12.5 An operational exACTiom for the framework

The source uses *EXACTiom* to seek a formal foundation. In this edition, the name is retained for the following explicit construction rather than for an unstated universal conservation law. [U1, pp. 35–40].

The hinge-field exACTiom

A consequential event is represented by a specified state trajectory reaching an eligible event set, under defined dynamics and a defined metric or residual, followed by a stated change in constraints, state, connectivity or governing equations.

Written compactly,

$$\dot{\mathbf{z}} = \mathbf{f}_m(\mathbf{z}, \mathbf{u}, t), \quad (12.1)$$

$$0 = g_i(\mathbf{z}, t), \quad \text{with the event's direction and eligibility,} \quad (12.2)$$

$$(\mathbf{z}^+, m^+) = \mathcal{R}_i(\mathbf{z}^-, m^-, t_*). \quad (12.3)$$

When the event is geometric, its residual can be built from an SDF and the mechanism's kinematics. When it is pressure-, damage- or evidence-dependent, it needs the corresponding state and units. The event equations do not obtain those additional states by renaming distance.

The resulting claim is precise and useful: the same *form of investigation* can be applied across disciplines. The material laws, observations and successful parameterisations remain local to each system. This is the transdisciplinary integration the professional title names.

12.6 Final professional statement

Tom Klootwijk

Principal Transdisciplinary Technologist | Computational Naturalist

I investigate the mechanisms that shape behaviour at boundaries, transitions and failure points. I combine natural observation, geometry, dynamics, computation and creative experimentation to build models, tools and inventions. My hinge-field framework connects the location of a boundary to the forces, signals, histories and state changes that make an event consequential. I lead the technical path from observing a process to defining it, testing it and making it usable.

NL200678942 / 10-07-1990

Computational naturalism, cognition, and creative systems design.

Source-to-model crosswalk

A.1 Project source register

U1. Tom Klootwijk project dialogue, *lieveheersbeestjes engels.pdf*. User-supplied, 41 pages. This is a recorded dialogue containing the user's prompts and earlier AI responses. Pages 37–39 introduce signed distance as “root and stem”; pages 39–41 introduce the small O pinion and spatial–temporal hinge vocabulary. Page numbering here refers to this 41-page edition, not the earlier 29-page file.

U2. Tom Klootwijk project dialogue, *Ibit dither of doubt psychologically of choice locusT.pdf*. User-supplied, 35 pages. Pages 1–9 develop the T , ϕ , Ψ and Δ vocabulary. Later pages extend it to optics, signal-processing analogies and environmental claims. This edition uses that vocabulary as a starting point and defines each retained mathematical object explicitly.

U3. *Hinge Events: A Physical Atlas of Boundaries, Capture, Discharge and Transformation*. Earlier 54-page project report supplied in the conversation. It provides continuity for the mechanism-atlas format. Empirical statements in the present edition are tied to primary research rather than treating the earlier report as a new experiment.

U4. Professional identity and integration request. Current user-supplied text adopting Principal Transdisciplinary Technologist and Computational Naturalist, together with the identity record Tom Klootwijk / NL200678942 / 10-07-1990. It provides the basis for the drafted professional and executive presentation.

A.2 What is retained, specified or changed

Root and stem. Retained as the organising relationship between geometric representation and the mechanism built upon it. An SDF supplies a location and orientation; dynamics and material laws supply the event's physical development.

Negative, zero and positive. Retained as inside, boundary and outside relative to a selected region. The source's assignment of those signs to symbolic categories is not used as a physical measurement. A geometrical sign has no such category meaning until a separate model defines one.

The Eikonal relation. Retained as the unit-gradient property of a genuine Euclidean distance field where differentiable. The source's claim that it forces equal-and-opposite changes in biological and virtual volume is not a consequence of that equation. Chapter 3 provides an explicit sphere counterexample and a separate formula for actual boundary-driven volume change.

Small O pinion. Retained as a project name for investigating constraints, transmission and hinge-like transitions. Literal pinion or gear anatomy is identified only where anatomical evidence supports it, as in the *Issus* example. It is not assigned to every ladybird joint.

ϕ and ψ . Specified as an angular coordinate and, where useful, a temporal phase. The axis is a unit vector $\mathbf{e}$; an eigenvector requires a specified operator. The source calls the letters themselves eigenvector axes without supplying the operator. Chapter 4 supplies the rotation matrix and identifies its actual axial eigenvector.

The four-coordinate matrix. Specified as a homogeneous rigid transform acting on $(x, y, z, 1)$, while time parameterises the transform. A matrix acting on (x, y, z, t) is a different object and requires a declared

spacetime metric and interpretation. It does not automatically conserve energy or mix seconds with metres meaningfully.

Ψ and one-bit events. Retained as local mode indicators or event outputs. Continuous state, multiple modes and history remain in the model. The source does not provide anatomical evidence for a universal master agency bit, so no such variable is treated as an observed biological component.

Delta–delta. Specified by the variable being differentiated, the derivative order and the sample interval. A second derivative of position is acceleration; jerk is the third derivative. Chapter 7 also derives the associated measurement-noise amplification.

The “5G noise register” and optical LUT. U2 contains software-like descriptions and assembly-style text, but does not supply a measured register, identified sensory transfer function or demonstrated causal path for those claims. They are not used as biological components in this edition. The engineered examples instead define actual model inputs, states, timing and computational functions.

Universal exactness. Applied here to stated mathematical constructions and their derivations. A calculation’s agreement with its analytic benchmark is different from a model’s agreement with an observed organism. The source terminology is preserved where it supports an explicit object; any additional physical mechanism has a research source or is named as a model developed here.

A reusable hinge dossier

B.1 The record to complete for a new mechanism

Field	Required content
Object and state	Species or engineered object, developmental or operating condition, specimen or model identifier.
Observed process	What was actually recorded, by which instrument and at what temporal and spatial resolution.
Geometry	Region Ω , boundary Γ , sign convention, coordinate frame, reconstruction method and geometric error.
Field	Exact distance, approximate distance, general implicit function, state margin or arrival time; include units.
Configuration	Joint coordinates, poses, constraints, limits and the maps from local geometry to world geometry.
Continuous state	Positions, velocities, temperature, concentration, pressure, damage, memory or other variables needed to predict change.
Coupling	Force, impulse, flux, reaction, receptor or control law connecting the interacting components.
Event	Residual g_i , crossing direction, eligibility, detection method and treatment of grazing or simultaneous conditions.
Consequence	Reset, new contact mode, change of connectivity, added or removed constraint, or changed governing equation.
History	Which variables retain past exposure, damage, deformation or input, and how they evolve.
Uncertainty	Parameter and measurement uncertainties, correlations, numerical error and the resulting event sensitivity.
Verification	Analytic or numerical reference case that tests the implementation independently of the observation.
Validation	Observation not used to set the parameters; prediction residuals and a competing model or alternative mechanism.
Technical output	Model, instrument, visualisation, prototype or invention brief produced from the completed investigation.

B.2 A compact review question

At the claimed hinge, what changes: the position, the contact relation, the admissible motion, the transferred material, the stored state, the governing equation, or only the label?

The answer determines which part of the framework is needed. It also prevents an attractive description of a boundary from replacing the mechanism that makes crossing it consequential.

Research and technical references

- [1] Osher, S., and Sethian, J. A. (1988). Fronts propagating with curvature-dependent speed: Algorithms based on Hamilton–Jacobi formulations. *Journal of Computational Physics*, 79, 12–49. DOI: 10.1016/0021-9991(88)90002-2. Author-hosted paper.
- [2] Hart, J. C. (1996). Sphere tracing: A geometric method for the antialiased ray tracing of implicit surfaces. *The Visual Computer*, 12, 527–545. DOI: 10.1007/s003710050084.
- [3] Lynch, K. M., and Park, F. C. (2017). *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press. Rotation discussion consulted through the authors’ official companion: exponential coordinates of rotation.
- [4] Sussman, M., Smereka, P., and Osher, S. (1994). A level set approach for computing solutions to incompressible two-phase flow. *Journal of Computational Physics*, 114, 146–159. DOI: 10.1006/jcph.1994.1155.
- [5] Tedrake, R. *Underactuated Robotics*, chapter “Multi-Body Dynamics.” Massachusetts Institute of Technology, online course text, accessed 17 September 2026. Official chapter.
- [6] Kong, N. J., Payne, J. J., Zhu, J., and Johnson, A. M. (2024). Saltation matrices: The essential tool for linearizing hybrid dynamical systems. *Proceedings of the IEEE*, 112(6), 585–608. DOI: 10.1109/JPROC.2024.3440211. Author manuscript, version 3.
- [7] Sethian, J. A. (1996). A fast marching level set method for monotonically advancing fronts. *Proceedings of the National Academy of Sciences*, 93, 1591–1595. DOI: 10.1073/pnas.93.4.1591. Author-hosted manuscript.
- [8] Burrows, M., and Sutton, G. (2013). Interacting gears synchronize propulsive leg movements in a jumping insect. *Science*, 341, 1254–1256. DOI: 10.1126/science.1240284. AAAS annotated primary paper.
- [9] Patek, S. N., Baio, J. E., Fisher, B. L., and Suarez, A. V. (2006). Multifunctionality and mechanical origins: Ballistic jaw propulsion in trap-jaw ants. *Proceedings of the National Academy of Sciences*, 103, 12787–12792. DOI: 10.1073/pnas.0604290103.
- [10] Arndt, E. M., Moore, W., Lee, W.-K., and Ortiz, C. (2015). Mechanistic origins of bombardier beetle (Brachinini) explosion-induced defensive spray pulsation. *Science*, 348, 563–567. DOI: 10.1126/science.1261166.
- [11] Challita, E. J., Sehgal, P., Krugner, R., and Bhamla, M. S. (2023). Droplet superpropulsion in an energetically constrained insect. *Nature Communications*, 14, 860. DOI: 10.1038/s41467-023-36376-5.
- [12] Van Wassenbergh, S., Ortlieb, E. J., Mielke, M., Böhmer, C., Shadwick, R. E., and Abourachid, A. (2022). Woodpeckers minimize cranial absorption of shocks. *Current Biology*, 32, 3189–3194.e4. DOI: 10.1016/j.cub.2022.05.052.
- [13] Meyer-Rochow, V. B., and Gal, J. (2003). Pressures produced when penguins pooh—calculations on avian defaecation. *Polar Biology*, 27, 56–58. DOI: 10.1007/s00300-003-0563-3.

- [14] Lovas-Kiss, Á., et al. (2020). Experimental evidence of dispersal of invasive cyprinid eggs inside migratory waterfowl. *Proceedings of the National Academy of Sciences*, 117, 15397–15399. DOI: 10.1073/pnas.2004805117.
- [15] Croft, B., et al. (2016). Contribution of Arctic seabird-colony ammonia to atmospheric particles and cloud-albedo radiative effect. *Nature Communications*, 7, 13444. DOI: 10.1038/ncomms13444.
- [16] Fredericksen, M. A., et al. (2017). Three-dimensional visualization and a deep-learning model reveal complex fungal parasite networks in behaviorally manipulated ants. *Proceedings of the National Academy of Sciences*, 114, 12590–12595. DOI: 10.1073/pnas.1711673114.
- [17] Turlings, T. C. J., Tumlinson, J. H., and Lewis, W. J. (1990). Exploitation of herbivore-induced plant odors by host-seeking parasitic wasps. *Science*, 250, 1251–1253. DOI: 10.1126/science.250.4985.1251.
- [18] Feynman, R. P., Leighton, R. B., and Sands, M. (1963). *The Feynman Lectures on Physics*, Volume I, chapter 31, “The Origin of the Refractive Index.” Caltech online edition.
- [19] Oleynikova, H., Taylor, Z., Fehr, M., Nieto, J., and Siegwart, R. (2017). Voxblox: Incremental 3D Euclidean signed distance fields for on-board MAV planning. *IEEE/RSJ International Conference on Intelligent Robots and Systems*. Author manuscript: arXiv 1611.03631.
- [20] Park, J. J., Florence, P., Straub, J., Newcombe, R., and Lovegrove, S. (2019). DeepSDF: Learning continuous signed distance functions for shape representation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. Author manuscript: arXiv 1901.05103.

Locate the boundary.
Specify the coupling.
Resolve the event.
Test what follows.

Tom Klootwijk

Principal Transdisciplinary Technologist

Computational Naturalist

BEATIFIC XRfysSion / HINGE FIELDS

Executive identity and a calculus of consequential boundaries.